

GEN- PROBE INCORPORATED

PANTHER Dashboard User Guide

Proprietary Information

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PANTHER[®] Remote Dashboard

User Guide

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Revision History

Date of Change	Revision	Description	Primary Contributor
2012.06.05	A	New	Cindy Salazar Michael Million

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Introduction

The *PANTHER Remote Dashboard* (“Dashboard”) is a software application that provides instrument runtime diagnostics, analytics, and reports to allow Field Service Engineers (“FSEs”) and technicians the ability to analyze instrument health, accuracy, and test results. The application is composed of individual modules (or mini applications) that allow a FSE to analyze and diagnose various system errors, warnings or test anomalies that may have occurred during an Assay procedure. The application supports compatibility across all PANTHER releases to date which include SPx (Diagnostics code branch) and Blood Bank/Real-Time code branch.

Purpose

This guide describes the Dashboard application functionality. This application resides on the PANTHER instrument and allows Gen-Probe Engineering, Gen-Probe FSEs/ISEs, and Gen-Probe partners to access instrument information that can be used to determine its functioning state. The Dashboard application allows Gen-Probe FSEs to maintain more PANTHER instruments since the travel time is removed. In addition, it allows Gen-Probe FSEs to properly identify the root cause to an issue, allowing a FSE to change only the necessary parts that are defective and reduce maintenance times.

Dashboard Modules

The FSEs are able to access PANTHER software data allowing quick diagnosis by providing the following high level functionality:

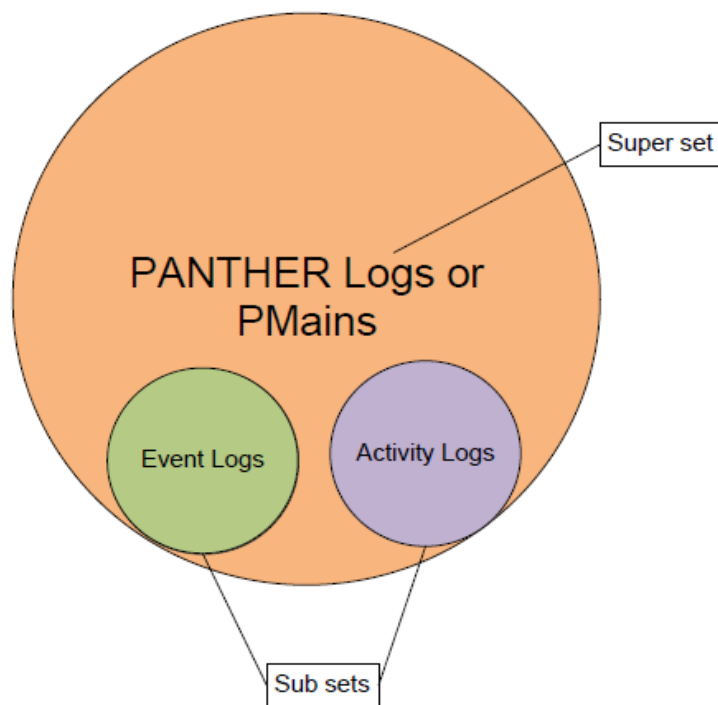
- Event Log view with sort and filter capabilities
- PANTHER Main Logs view with sort and filter capabilities
- Worklist flag roll up display
- Test Order Flag detail display
- Test Order Process Control information graphing, e.g. pressure curves
- Flash Check run evaluation
- Sys Check run evaluation
- HPV PQ run evaluation
- AC2 PQ run evaluation
- LIS Log Continuous Display
- PRO360° Networking Troubleshooting tools
- Reporting
 - PANTHER Report

- o One Day Health Report
- o System Health Report
- o Manufacturing Health Report
- o Flash Check and Sys Check Analysis Reports

The following sections describe the Dashboard modules in a great level of detail...

EVENT LOGS

The Dashboard *Event Logs* module provides the ability to view and analyze PANTHER System Log files. The overall PANTHER System Log files can be referred to as '*PMains*' and are considered the super set while the '*Event Logs*' and '*Activity Logs*' are considered the sub sets.



A *PANTHER Log* file is a recording of important user and application activity that occurs on the instrument. Each entry in the PANTHER Log contains a date/time stamp. This allows the information to be chronologically ordered to assist with problem troubleshooting. If something should malfunction within the complex PANTHER system, there may be no other way of identifying the problem so that when the problem occurs it is easier to pinpoint and fix.

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The *Event Logs* module displays *Event Log* and *Activity Log* information which includes PANTHER instrument diagnostic messages as well as their severity levels. Main Log entries that correspond to the same time frame as a selected *Event Log* entry can be viewed. Once the Events and or Activity records are populated in the Event and/ or Activity grid, the user can select an Event or Activity record and generate a detailed log file search to get the *PANTHER Log*, or *PMains*, detailed information related to the selected entries.

Event Logs and *Activity Logs* are displayed in the upper section of the screen while the *PANTHER Logs*, or *PMains*, are displayed at the bottom of the screen.

To open a *PMains* file, click on the “*Record Creation Time*” column and the corresponding *PMains* file will display at the bottom of the screen.

Event Logs Flags by Worklist Graphing Health Reports Flash Check Sys Check Quantitative Graphing AC2 PQ HPV PQ LIS Monitor PRO 360 Connectivity Tester								
Drag a column here to group by this column.								
Record Creation Time	Event ID	Description	Type	Event #	Severity	Module	Source	
09/23/2011 4:17:10.527	032.001.00113	Incubator temperature is now in range	EventLogEntry	113	Message	HIGH TEMP INCUBATOR1	Incubator1	
09/23/2011 4:14:59.520	032.001.00112	Incubator1 temperature is out of range	EventLogEntry	112	Warning	HIGH TEMP INCUBATOR1	Incubator1	
09/23/2011 4:24:04.100	048.000.00047	Dwell time is ended after	EventLogEntry	47	Message	MAGNETIC WASH STATION1	MagWash1	
09/23/2011 4:24:04.077	048.000.00047	Dwell time is ended after	EventLogEntry	47	Message	MAGNETIC WASH STATION1	MagWash1	
09/23/2011 4:24:02.053	048.000.00046	Dwell time is started for	EventLogEntry	46	Message	MAGNETIC WASH STATION1	MagWash1	
09/23/2011 4:24:02.020	048.000.00046	Dwell time is started for	EventLogEntry	46	Message	MAGNETIC WASH STATION1	MagWash1	
09/23/2011 4:23:19.540	048.000.00047	Dwell time is ended after	EventLogEntry	47	Message	MAGNETIC WASH STATION1	MagWash1	
09/23/2011 4:23:19.510	048.000.00047	Dwell time is ended after	EventLogEntry	47	Message	MAGNETIC WASH STATION1	MagWash1	
09/23/2011 4:23:17.497	048.000.00046	Dwell time is started for	EventLogEntry	46	Message	MAGNETIC WASH STATION1	MagWash1	
09/23/2011 4:23:17.467	048.000.00046	Dwell time is started for	EventLogEntry	46	Message	MAGNETIC WASH STATION1	MagWash1	
09/23/2011 4:24:04.157	049.000.00047	Dwell time is ended after	EventLogEntry	47	Message	MAGNETIC WASH STATION2	MagWash2	
09/23/2011 4:24:04.127	049.000.00047	Dwell time is ended after	EventLogEntry	47	Message	MAGNETIC WASH STATION2	MagWash2	
09/23/2011 4:24:02.103	049.000.00046	Dwell time is started for	EventLogEntry	46	Message	MAGNETIC WASH STATION2	MagWash2	
09/23/2011 4:24:02.077	049.000.00046	Dwell time is started for	EventLogEntry	46	Message	MAGNETIC WASH STATION2	MagWash2	
09/23/2011 4:23:19.600	049.000.00047	Dwell time is ended after	EventLogEntry	47	Message	MAGNETIC WASH STATION2	MagWash2	
09/23/2011 4:23:19.560	049.000.00047	Dwell time is ended after	EventLogEntry	47	Message	MAGNETIC WASH STATION2	MagWash2	
09/23/2011 4:23:17.570	049.000.00046	Dwell time is started for	EventLogEntry	46	Message	MAGNETIC WASH STATION2	MagWash2	
09/23/2011 4:23:17.523	049.000.00046	Dwell time is started for	EventLogEntry	46	Message	MAGNETIC WASH STATION2	MagWash2	
09/23/2011 4:25:10.000	407.009.00001	System state entered: Setup	ActivityLogEntry	1	Message	PantherInterface	PantherInterface	
09/23/2011 4:00:19.107	407.009.00001	System state entered: Maintenance	ActivityLogEntry	1	Message	PantherInterface	PantherInterface	
09/22/2011 19:19:33.833	407.009.00001	System state entered: Setup	ActivityLogEntry	1	Message	PantherInterface	PantherInterface	
09/22/2011 17:10:50.217	407.009.00001	System state entered: Ready	ActivityLogEntry	1	Message	PantherInterface	PantherInterface	
09/22/2011 15:26:06.533	407.007.00002	MTUs loaded	ActivityLogEntry	2	Message	PantherInterface	PantherInterface	
09/22/2011 14:47:42.300	407.006.00002	Rack Unloaded: 001351	ActivityLogEntry	2	Message	PantherInterface	PantherInterface	
09/22/2011 14:47:40.447	407.006.00002	Rack Unloaded: 000778	ActivityLogEntry	2	Message	PantherInterface	PantherInterface	
09/22/2011 14:47:39.700	407.006.00002	Rack Unloaded: 000974	ActivityLogEntry	2	Message	PantherInterface	PantherInterface	
09/22/2011 14:47:39.020	407.006.00002	Rack Unloaded: 001425	ActivityLogEntry	2	Message	PantherInterface	PantherInterface	
09/22/2011 12:57:32.413	407.006.00002	Rack Unloaded: 001368	ActivityLogEntry	2	Message	PantherInterface	PantherInterface	
09/22/2011 12:57:28.950	407.006.00002	Rack Unloaded: 000634	ActivityLogEntry	2	Message	PantherInterface	PantherInterface	
09/22/2011 12:57:26.760	407.006.00002	Rack Unloaded: 001356	ActivityLogEntry	2	Message	PantherInterface	PantherInterface	
09/22/2011 12:18:22.683	407.007.00002	MTUs loaded	ActivityLogEntry	2	Message	PantherInterface	PantherInterface	

Event Logs – Module Components

Column Headers

Record Creation Time – the time the record was created. It is displayed by local date, hour, minute, seconds and milliseconds.

EventID – event identification number.

Description – describes the message details for *Event Log(s)* or *Activity Log(s)*.

Type – lists the type of log, i.e. *Event Log* or *PANTHER Log*.

Event # – event number.

Severity – lists the severity of a log, i.e. Message, Warning, Problem, Critical, or Fatal.

Module – PANTHER component name.

Source – PANTHER source name.

Event Logs Functionality

- 1) Event Log messages can be viewed for each hardware module.
- 2) A maximum of 500 Event Log records can be displayed.
- 3) Event Log sorting, conditional formatting, and grouping can be done by right-clicking the column header.

Event Logs		Flags by Worklist	Graphing	Health Reports	Flash Check	Sys Check	Quantitative Graphing	AC2 PQ	HPV
Drag a column here to group by this column.									
	Record Creation Time	Event ID	Description	Type					
	09/20/2011 9:03:54.587	407.005.00001	Waste Inventory started						
	09/16/2011 8:54:51.940	407.005.00001	Waste Inventory started						
	09/13/2011 11:02:19.937	407.005.00001	Waste Inventory started						
	09/14/2011 16:07:40.463	407.002.00001	Valid Assay Reagent Kit Loaded: SysCheck						
	09/20/2011 12:35:30.837	407.002.00001	Valid Assay Reagent Kit Loaded: CT/GC						
	09/19/2011 13:04:48.860	407.002.00001	Valid Assay Reagent Kit Loaded: CT/GC						
	09/16/2011 12:09:03.273	407.002.00001	Valid Assay Reagent Kit Loaded: CT/GC						
	09/15/2011 11:43:51.077	407.002.00001	Valid Assay Reagent Kit Loaded: CT/GC						
	09/13/2011 11:44:23.257	407.002.00001	Valid Assay Reagent Kit Loaded: CT/GC						
	09/22/2011 11:26:32.190	407.002.00001	Valid Assay Reagent Kit Loaded:						
	09/23/2011 7:09:47.397	024.001.00067	Vacuum pump maximum speed reached						
	09/23/2011 4:48:34.607	024.001.00067	Vacuum pump maximum speed reached						

Sort Ascending

☒ Sort Descending

Clear Sorting

Conditional Formatting

Group by this column

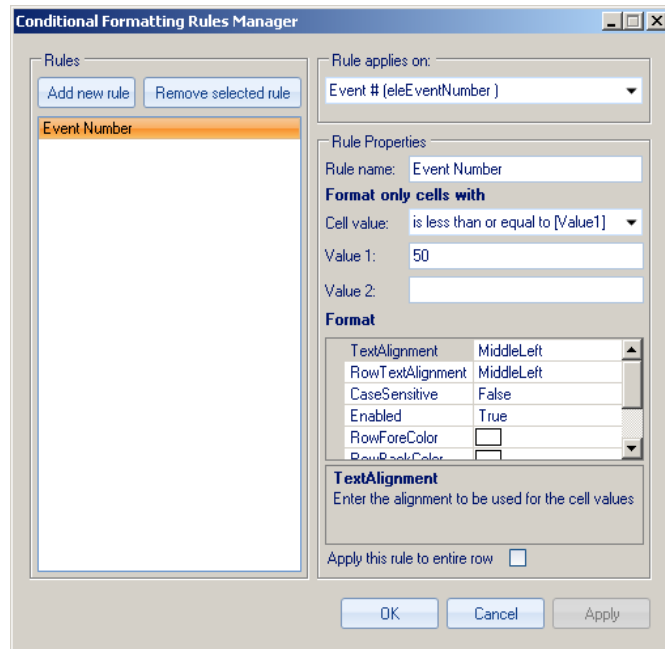
Column Chooser

Hide Column

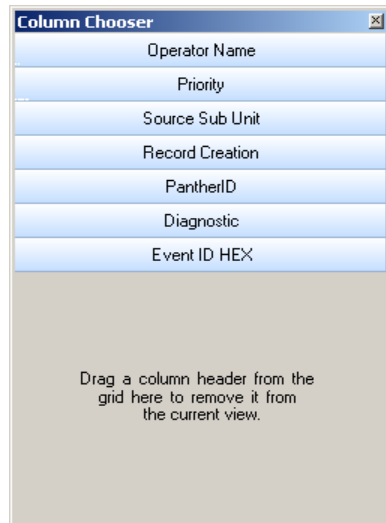
Pin Column

Best Fit

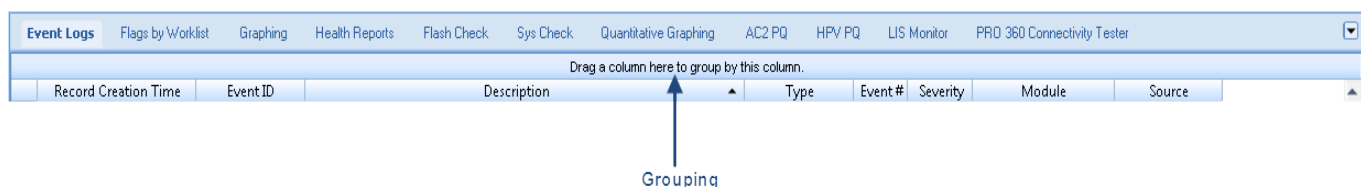
- A) Columns can be sorted in ascending or descending order by right-clicking on the column header and selecting 'Sort Ascending' or 'Sort Descending.'
- B) Column sorting can be reversed by right-clicking on the column header and selecting 'Clear Sorting.'
- C) Conditional Formatting can be done by right-clicking on the column header and selecting 'Conditional Formatting.' This feature assists with the creation of rules for customized formatting. The following can be controlled: text alignment, row alignment, case sensitive, row color, and cell color.



- D) Columns can be grouped by an existing column by right-clicking on the column header and selecting *'Group by this column.'* Grouping can be used to consolidate non-numeric data values into more general group values; these group values can be mapped into a new column within the data set.
- E) Columns can be removed from the current view by right-clicking on the column header and selecting *'Column Chooser.'* Drag a column header into the gray area in the *'Column Chooser'* dialog box to remove the column.



- F) Columns can be hidden from the *Event Logs* screen by right-clicking on the column header and selecting '*Hide Column.*' To unhide the column, right-click on any column header and select '*Column Chooser,*' then drag the hidden column header back out to the '*Event Logs*' screen.
 - G) Columns can be 'pinned' so that they remain at the ends of the screen (left or right side); this keeps the column headings visible as the data bar is scrolled up and down. To pin the column, right-click on the column header and select '*Pin Column;*' to reverse this, right-click on the same column header and select '*Unpin Column.*'
 - H) Columns do not automatically accommodate to fit the information they contain. However, by right-clicking and selecting '*Best Fit,*' the column automatically expands and contracts to fit the data enclosed.
- 4) *Event Logs* can also be grouped by holding the left-click mouse button down and dragging a column to the row named "*Drag a column here to group by this column.*"



- A) The *Event Logs* are grouped by '*Module*' and '*Severity.*'

Module	Severity	Record Creation Time	EventID	Description	Type	Event #	Source
▼ Module: HIGH TEMP INCUBATOR1;							
▼ Module: MAGNETIC WASH STATION1;							
▼ Module: MAGNETIC WASH STATION2;							
▼ Module: ManualActions;							
▼ Module: PantherInterface;							
▼ Module: PantherUI;							
▲ Module: ProcessManagement;							
▲ Severity: Message;							
		09/23/2011 4:14:59.553	259.000.00003	MTU introduction paused due to TemperatureOutOfRangeCritical.	EventLogEntry	3	ProcessManage...
		09/23/2011 4:17:10.560	259.000.00004	MTU introduction resumed.	EventLogEntry	4	ProcessManage...
▼ Module: TestOrderFactory;							
▼ Module: VACUUM COOLING;							

- 5) *Event Logs* can also be sorted/ filtered in reverse chronological order based on the creation time/ date by using the arrow  that appears to the right of the column header; click on the column header for the arrow to be displayed.

Description
Waste Inventory started
Waste Inventory started
Waste Inventory started
Valid Assay Reagent Kit Loaded: SysCheck
Valid Assay Reagent Kit Loaded: CT/GC
Valid Assay Reagent Kit Loaded: CT/GC
Valid Assay Reagent Kit Loaded: CT/GC
Valid Assay Reagent Kit Loaded: CT/GC
Valid Assay Reagent Kit Loaded: CT/GC
Valid Assay Reagent Kit Loaded:
Vacuum pump maximum speed reached
Vacuum pump maximum speed reached
Vacuum pump maximum speed reached
Vacuum pump maximum speed reached
Vacuum pump maximum speed reached
Vacuum pressure is out of range
Vacuum pressure is out of range
Vacuum pressure is out of range
Vacuum pressure back in range (653 C)
Vacuum pressure back in range (653 C)
Vacuum pressure back in range (652 C)
User shutdown
User shutdown
User shutdown
User logged on
User logged on

 Filtering Arrow

- 6) *Event Logs* can also be filtered by a text search; type the word in the text field under the column header to complete filtering.

Description
User
User shutdown
User shutdown
User shutdown
User logged on
User logged on
User logged on
User logged off
User logged off
User logged off

Text Search Filtering

- 7) *Event Log* columns can also be expanded or contracted by placing the cursor in between the columns, holding down the left-click button on the mouse, and dragging the bi-directional arrow in the desired direction.
- 8) *Event Log* columns can be moved and re-organized by placing the cursor on the column header, holding down the left-click button on the mouse, and dragging it into the desired location.
- 9) *Event Log* entries which have been viewed and bookmarked remain displayed upon returning to the *Event Log* screen.

Event Log Search

Search for *Event Logs* by selecting a date range and checking-off '*Severity*,' '*Log Type*,' and '*Include Diagnostic Messages*' as preferred.

Use the '*Description*' text box to type description notes.

PANTHER Log Search

Search for *PANTHER Logs* by date using the arrow/ drop-down and clicking on ‘Search.’

The ‘Clear’ button re-sets the ‘Search Date’ to the present date.

Results are displayed at the bottom of the *Event Logs* screen and are broken down by the following columns:

Date/Time – Local date and time of the PMains Log.

Content – describes the instrument message details.

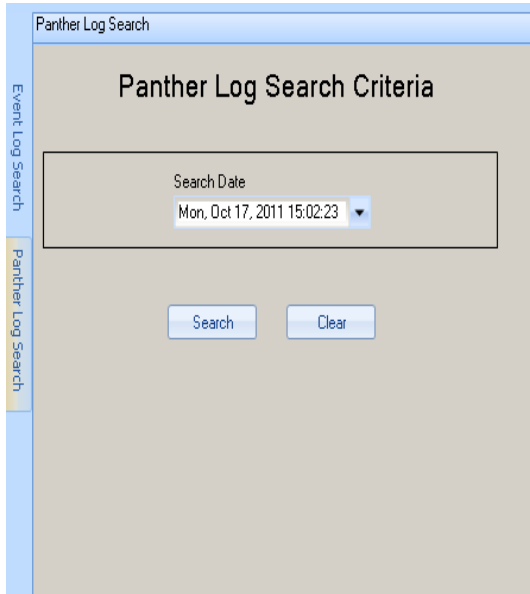
Category – classifies PMains information, e.g. Data Manager, PANTHER UI, RS232 Communication.

Error Level – describes the severity of the error.

Thread – thread number; it is PANTHER system generated.

Class Name – message class, e.g. Event Logs, Activity Logs.

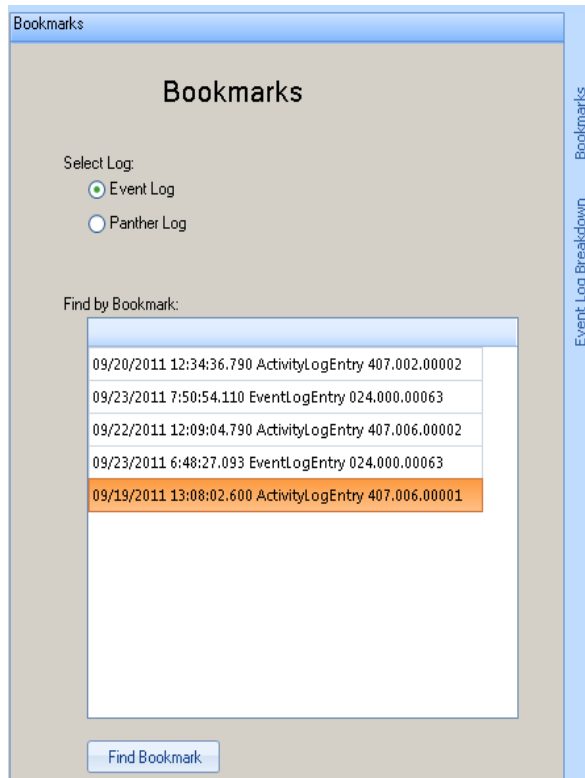
Log File – displays the PANTHER Log or *PMain* log file this entry derives from.



The screenshot displays a web application window titled "Panther Log Search". On the left side, there is a vertical navigation bar with three tabs: "Event Log Search", "Panther Log Search" (which is currently selected and highlighted in orange), and another "Panther Log Search" tab below it. The main content area of the window is titled "Panther Log Search Criteria". Within this area, there is a section labeled "Search Date" containing a dropdown menu that shows "Mon, Oct 17, 2011 15:02:23". Below the dropdown menu, there are two buttons: "Search" and "Clear".

Bookmarks

Book Marks can be set or reset on any of the detailed grid log entries to allow the user to return to the selected locations during the log file evaluations. A current *Event Log* or *PANTHER log* entry can be marked in order to easily return to it by using the '*Bookmarks*' tab.



Setting Bookmarks – from the ‘Events Log’ screen, select an *Event Log* or *PANTHER Log* by clicking on a desired log row, then right-click and select ‘Set Bookmark.’

Displaying the Bookmarks list – click on the ‘Bookmarks’ tab and select ‘Event Log’ or ‘PANTHER Log’ to see the appropriate list.

Finding a Bookmark from the Logs list – from the ‘Bookmarks’ tab, select the desired log, and click on the ‘Find Bookmark’ button.

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Event Logs

Flags by Worklist

Graphing

Health Reports

Flash Check

Sys Check

Quantitative Graphing

AC2 PQ

HPV PQ

L

Drag a column here to group by this column.

Record Creation Time	Severity	Event ID	Description	Type
09/14/2011 8:46:58.743	Message	407.013.00004	Alarm silenced	ActivityLogEn
09/22/2011 11:25:24.170	Message	407.002.00002	Assay Reagent Kit Unloaded: CT/GC	ActivityLogEn
09/20/2011 12:34:36.790	Message	407.002.00002	Assay Reagent Kit Unloaded: CT/GC	ActivityLogEn
09/19/2011 11:50:51.797	Message	407.002.00002	Assay Reagent Kit Unloaded: CT/GC	ActivityLogEn
09/16/2011 12:08:17.030	Message	407.002.00002	Assay Reagent Kit Unloaded: CT/GC	ActivityLogEn
09/16/2011 8:43:34.943	Message	407.002.00002	Assay Reagent Kit Unloaded: CT/GC	ActivityLogEn
09/15/2011 11:43:10.610	Message	407.002.00002	Assay Reagent Kit Unloaded: CT/GC	ActivityLogEn
09/14/2011 10:57:52.057	Message	407.002.00002	Assay Reagent Kit Unloaded: CT/GC	ActivityLogEn
09/13/2011 10:53:04.647	Message	407.002.00002	Assay Reagent Kit Unloaded: CT/GC	ActivityLogEn
09/22/2011 10:43:12.600	Message	407.013.00002	Cleared all messages	ActivityLogEn
09/19/2011 11:49:15.767	Message	407.013.00002	Cleared all messages	ActivityLogEn
09/23/2011 7:51:03.837	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:50:54.110	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:50:50.010	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:50:44.890	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:50:43.353	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:50:40.280	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:50:27.500	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:50:00.350	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:34:56.627	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:34:55.093	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:34:53.550	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:34:51.500	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:34:05.933	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:18:22.270	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:18:20.223	Message	024.000.00063	Cooling liquid too cold	EventLogEntr
09/23/2011 7:18:19.227	Message	024.000.00063	Cooling liquid too cold	EventLogEntr

Bookmarks

Select Log:
☒ Event Log
☐ Panther Log

Find by Bookmark:

09/20/2011 12:34:36.790 ActivityLogEntry 407.002.00002

09/23/2011 7:50:54.110 EventLogEntry 024.000.00063

Find Bookmark

Event Log Breakdown

Event Log Breakdown

This tab is located on the right side of the 'Event Logs' screen. Logs are displayed by individual module with the amount of messages, warnings, problems, critical, fatal and unknown frequencies.

Event Log Breakdown							
	Module Name	Mes...	War...	Prob...	Criti...	Fatal...	Unk...
	HIGH TEMP INCUBATO...	1	1	0	0	0	0
	ProcessManagement	2	0	0	0	0	0
	MAGNETIC WASH STA...	8	0	0	0	0	0
	MAGNETIC WASH STA...	8	0	0	0	0	0
	PantherUI	9	0	0	0	0	0
	PantherInterface	87	0	0	0	0	0
	TestOrderFactory	98	0	0	0	0	0
►	VACUUM COOLING	474	0	0	0	3	0

FLAGS BY WORK LIST

The *Flags by Worklist* module provides the ability to analyze test *flags* from *Test Order* within a set of work lists. The work lists are retrieved from the DM database and displayed in a master grid by providing a start-date and end-date range. Overall, this module determines the number of pipetted tests per work list by counting the number of sample replicates aspirated. It also determines the number of MTUs processed per work list from the PANTHER test order table.

File About

Event Logs **Flags by Worklist** Graphing Health Reports Flash Check Sys Check Quantitative Graphing AC2 PQ HPV

Work List Filter Criteria

Start Date Friday, August 27, 2010

End Date Friday, October 21, 2011

Search Reset

Flags By Work List

Start	End	Work List ID	Duration	Pipetted Tests	MTUs Processed	Flags
9/22/2011 6:28:27 PM	9/23/2011 12:10:49 AM	00081-20110922-01	05:42:22	100	21	DUO<100>;
9/20/2011 8:17:08 PM	9/21/2011 8:10:16 PM	00081-20110920-01	23:53:08	250	50	DUO<250>; NTI<2>; RDFS<2>; p<2>;
9/19/2011 8:10:13 PM	9/20/2011 3:37:29 AM	00081-20110919-01	07:27:16	250	50	DUO<250>; NTI<2>; RDFS<2>; p<2>;
9/16/2011 7:17:33 PM	9/17/2011 2:44:49 AM	00081-20110916-01	07:27:16	250	50	DUO<250>;
9/15/2011 6:53:45 PM	9/16/2011 2:21:00 AM	00081-20110915-01	07:27:14	250	50	DUO<250>;
9/13/2011 6:48:27 PM	9/14/2011 2:20:43 AM	00081-20110913-01	07:32:15	250	50	DUO<250>;
9/12/2011 6:12:07 PM	9/13/2011 1:24:22 AM	00081-20110912-01	07:12:15	235	47	DUO<235>;
9/9/2011 7:36:14 PM	9/10/2011 1:48:30 AM	00081-20110909-01	06:12:15	152	31	DUO<152>;

Flags By Work List – Module Components

Column Headers

Start – local date and time that the work list began to run.

End – local date and time that the work list was completed.

Work List ID – work list identification.

Duration – total time it took to complete a work list.

Pipetted Tests – number of pipette tests completed per work list.

MTUs Processed – number of MTUs/ Multi-Tube Units processed per work list; 1 MTU = 5 disposable reaction tubes.

Flags – an error message.

Flags By Work List – Functionality

Result and Assay Flags can be displayed by entering a date range and selecting a start date, an end date, and then clicking on the 'Search' button.

The 'Reset' button adjusts the 'Start Date' and 'End Date' to the past seven days, ending with the present date.

Result and Assay Flags can be sorted by the 'Start,' 'End,' 'Work List ID,' 'Duration,' 'Pipetted Tests,' and 'MTUs Processed' by using the arrow  that appears to the right of the column header; click on the column header for the arrow to be displayed.

Start	End	Work List ID	Duration	Pipetted Tests	MTUs Processed	Flags
9/22/2011 6:28:27 PM	9/23/2011 12:10:49 AM	00081-2011092...	05:42:22	100	21	DUO<100>;
9/9/2011 7:36:14 PM	9/10/2011 1:48:30 AM	00081-2011090...	06:12:15	152	31	DUO<152>;
9/12/2011 6:12:07 PM	9/13/2011 1:24:22 AM	00081-2011091...	07:12:15	235	47	DUO<235>;
9/13/2011 6:48:27 PM	9/14/2011 2:20:43 AM	00081-2011091...	07:32:15	250	50	DUO<250>;
9/15/2011 6:53:45 PM	9/16/2011 2:21:00 AM	00081-2011091...	07:27:14	250	50	DUO<250>;
9/16/2011 7:17:33 PM	9/17/2011 2:44:49 AM	00081-2011091...	07:27:16	250	50	DUO<250>;
9/19/2011 8:10:13 PM	9/20/2011 3:37:29 AM	00081-2011091...	07:27:16	250	50	DUO<250>; NTI<2>; RDFS<2>; p<2>;
9/20/2011 8:17:08 PM	9/21/2011 8:10:16 PM	00081-2011092...	23:53:08	250	50	DUO<250>; NTI<2>; RDFS<2>; p<2>;

Filtering Arrow

Result and Assay Flags copying, sorting, conditional formatting, and grouping can also be done by right-clicking the column header.

Start	End	Work List ID	Duration	Pipetted Tests	MTUs Processed	Flags
9/22/2011 6:28:27 PM	9/23/2011 12:10:49 AM	00081-20110922-01	05:42:22	100	21	DUO<100>;
9/20/2011 8:17:08 PM	9/21/2011 8:10:16 PM	00081-20110920-01			50	DUO<250>; NTI<2>; RDFS<2>; p<2>;
9/19/2011 8:10:13 PM	9/20/2011 3:37:29 AM	00081-20110919-01			50	DUO<250>; NTI<2>; RDFS<2>; p<2>;
9/16/2011 7:17:33 PM	9/17/2011 2:44:49 AM	00081-20110916-01			50	DUO<250>;
9/15/2011 6:53:45 PM	9/16/2011 2:21:00 AM	00081-20110915-01			50	DUO<250>;
9/13/2011 6:48:27 PM	9/14/2011 2:20:43 AM	00081-20110913-01			50	DUO<250>;
9/12/2011 6:12:07 PM	9/13/2011 1:24:22 AM	00081-20110912-01			47	DUO<235>;
9/9/2011 7:36:14 PM	9/10/2011 1:48:30 AM	00081-20110909-01			31	DUO<152>;

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Result and Assay Flag columns can be expanded by placing the cursor in between the columns, holding down the left-click button on the mouse, and dragging the bi-directional arrow in the desired direction.

Result and Assay Flag columns can be moved and re-organized by placing the cursor on the column header, holding down the left-click button on the mouse, and dragging it into the desired location.

Test Order information can be viewed by selecting the Work List ID row.

File About

Event Logs **Flags by Worklist** Graphing Health Reports Flash Check Sys Check Quantitative Graphing AC2 PQ HPV PQ LIS Monitor PRO 360 Connectivity Tester

Work List Filter Criteria

Start Date: Monday, August 30, 2010

End Date: Monday, October 17, 2011

Search Reset

Flags By Work List

Start	End	Work List ID	Duration	Pipetted Tests	MTUs Processed	Flags
9/22/2011 6:28:27 PM	9/23/2011 12:10:49 AM	00081-2011...	05:42:22	100	21	DUO<100>;
9/20/2011 8:17:08 PM	9/21/2011 8:10:16 PM	00081-2011...	23:53:08	250	50	DUO<250>; NTI<2>; RDFS<2>; p<2>;
9/19/2011 8:10:13 PM	9/20/2011 3:37:29 AM	00081-2011...	07:27:16	250	50	DUO<250>; NTI<2>; RDFS<2>; p<2>;
9/16/2011 7:17:33 PM	9/17/2011 2:44:49 AM	00081-2011...	07:27:16	250	50	DUO<250>;
9/15/2011 6:53:45 PM	9/16/2011 2:21:00 AM	00081-2011...	07:27:14	250	50	DUO<250>;
9/13/2011 6:48:27 PM	9/14/2011 2:20:43 AM	00081-2011...	07:32:15	250	50	DUO<250>;
9/12/2011 6:12:07 PM	9/13/2011 1:24:22 AM	00081-2011...	07:12:15	235	47	DUO<235>;
9/9/2011 7:36:14 PM	9/10/2011 1:48:30 AM	00081-2011...	06:12:15	152	31	DUO<152>;

Test Order Information

Tube #	Completion	MTU...	MTU ID	RLU	Flags
1	9/19/2011 11:32:30 P...	1	1085917508	613	DUO;
2	9/19/2011 11:32:30 P...	2	1085917508	1040	DUO;
3	9/19/2011 11:32:30 P...	3	1085917508	5	DUO;
4	9/19/2011 11:32:30 P...	4	1085917508	7	DUO;
5	9/19/2011 11:32:30 P...	5	1085917508	5	DUO;
6	9/19/2011 11:37:30 P...	1	1085917510	4	DUO;
7	9/19/2011 11:37:30 P...	2	1085917510	5	DUO;
8	9/19/2011 11:37:30 P...	3	1085917510	5	DUO;
9	9/19/2011 11:37:30 P...	4	1085917510	997	DUO;
10	9/19/2011 11:37:30 P...	5	1085917510	922	DUO;
11	9/19/2011 11:42:30 P...	1	1085917517	981	DUO;
12	9/19/2011 11:42:30 P...	2	1085917517	1123	DUO;
13	9/19/2011 11:42:30 P...	3	1085917517	1132	DUO;
14	9/19/2011 11:42:30 P...	4	1085917517	1107	DUO;
15	9/19/2011 11:42:30 P...	5	1085917517	2057	DUO;
16	9/19/2011 11:47:30 P...	1	1085878052	2115	DUO;
17	9/19/2011 11:47:30 P...	2	1085878052	2116	DUO;
18	9/19/2011 11:47:30 P...	3	1085878052	811	DUO;
19	9/19/2011 11:47:30 P...	4	1085878052	703	DUO;
20	9/19/2011 11:47:30 P...	5	1085878052	379	DUO;
21	9/19/2011 11:52:30 P...	1	1085878053	811	DUO;
22	9/19/2011 11:52:30 P...	2	1085878053	871	DUO;
23	9/19/2011 11:52:30 P...	3	1085878053	810	DUO;
24	9/19/2011 11:52:30 P...	4	1085878053	7	DUO;
25	9/19/2011 11:52:30 P...	5	1085878053	5	DUO;
26	9/19/2011 11:57:30 P...	1	1085878054	5	DUO;
27	9/19/2011 11:57:30 P...	2	1085878054	5	DUO;
28	9/19/2011 11:57:30 P...	3	1085878054	5	DUO;
29	9/19/2011 11:57:30 P...	4	1085878054	5	DUO;
30	9/19/2011 11:57:30 P...	5	1085878054	819	DUO;
31	9/20/2011 12:02:30...	1	1085878055	979	DUO;
32	9/20/2011 12:02:30...	2	1085878055	1032	DUO;
33	9/20/2011 12:02:30...	3	1085878055	1134	DUO;
34	9/20/2011 12:02:30...	4	1085878055	1134	DUO;
35	9/20/2011 12:02:30...	5	1085878055	1139	DUO;
36	9/20/2011 12:07:30...	1	1085878056	2027	DUO;
37	9/20/2011 12:07:30...	2	1085878056	2099	DUO;
38	9/20/2011 12:07:30...	3	1085878056	2033	DUO;

Test Order Information – Column Headers

Tube # – tube number.

Completion – local date and time that the test order is completed.

MTU Pos – Multi-Tube Unit position number; 1 MTU = 5 disposable reaction tubes.

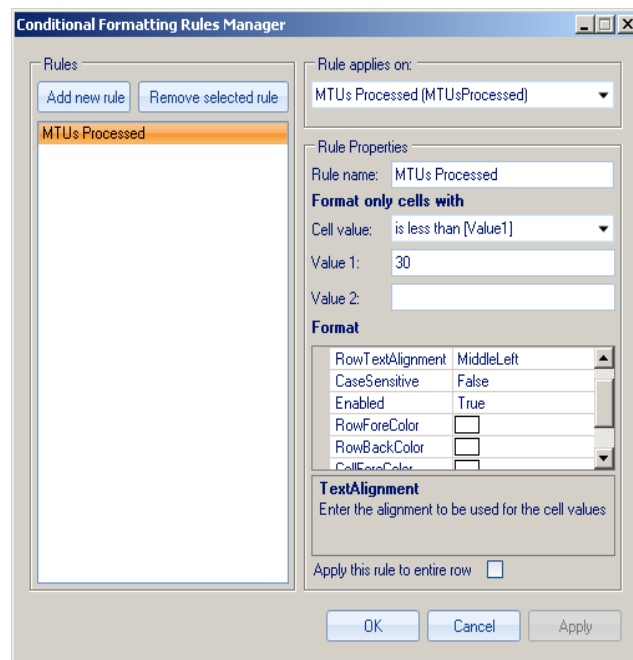
MTU ID – Multi-Tube Unit Identification number.

RLU – Relative Light Unit, it is a measurement used by Luminometers to quantify light emitted.

Flags – an error message.

Test Order Look Up columns can be organized by sorting, conditional formatting, and grouping by right-clicking the column header and selecting the listed functionality.

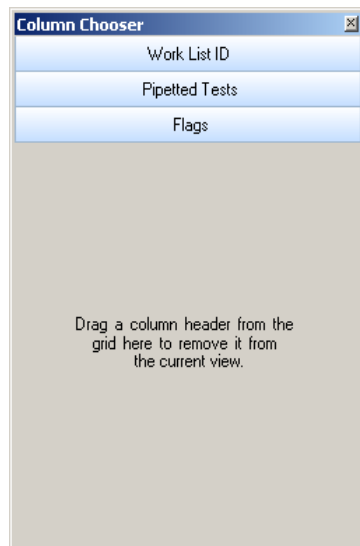
- A) Columns can be sorted in ascending or descending order by right-clicking on the column header and selecting 'Sort Ascending' or 'Sort Descending.'
- B) Column sorting can be reversed by right-clicking on the column header and selecting 'Clear Sorting.'
- C) Conditional Formatting can be done by right-clicking on the column header and selecting 'Conditional Formatting.' This feature assists with the creation of rules for customized formatting. The following can be controlled: text alignment, row alignment, case sensitive, row color, and cell color.



- D) Columns can be grouped by an existing column by right-clicking on the column header and selecting 'Group by this column.' Grouping can be used to consolidate non-numeric

data values into more general group values; these group values can be mapped into a new column within the data set.

- E) Columns can be removed from the current view by right-clicking on the column header and selecting '*Column Chooser.*' Drag a column header into the gray area in the '*Column Chooser*' dialog box to remove the column.



- F) Columns can be hidden from the Event Logs screen by right-clicking on the column header and selecting '*Hide Column.*' To unhide the column, right-click on any column header and select '*Column Chooser,*' then drag the hidden column header back out to the '*Event Logs*' screen.
- G) Columns can be 'pinned' so that they remain at the ends of the screen (left or right side); this keeps the column headings visible as the data bar is scrolled up and down. To pin the column, right-click on the column header and select '*Pin Column;*' to reverse this, right-click on the same column header and select '*Unpin Column.*'
- H) Columns do not automatically accommodate to fit the information they contain. However, by right-clicking and selecting '*Best Fit,*' the column automatically expands and contracts to fit the data enclosed.

GRAPHING

The graphing module allows the user the ability to view and analyze graphs based upon the Assay processing tests and command actions. This module can display information for up to 20,000 test

orders. The following graphs can be generated: Dispense, Aspiration, C-LLD, Post Disp Cap LLD, Shearing Pressure, Luminometer, Output Queue, Flash, MagWash, Pump Monitor, and Fluorescence.

Graphing – Functionality

Test Order Look Up – the Graphing module can plot MTU tube process control data from a local date/time range or a Worklist ID.

Search By Date Range

- 1) Click on the '*Search by Date Range*' button and select the start and end dates and times.
- 2) Click on the '*Search*' button to generate the test order list.

The '*Reset Dates*' button adjusts the '*Start Date*' and '*End Date*' to the past seven days, ending with the present date.

The test order columns can be expanded by placing the cursor in between the columns, holding down the left-click button on the mouse, and dragging the bi-directional arrow in the desired direction.

The test order columns can be moved and re-organized by placing the cursor on the column header, holding down the left-click button on the mouse, and dragging it into the desired location.

Search

Selections

Test Order Look Up

☒ Search By Date Range
 ☐ Search By WorkListID

Start
 End
 Work List ID

Drag a column here to group by this column.

Graph	Creation Time	Test Or...	Work List ID	Sample ID	MTU ID	Posit... ▲	Flags ▲
	9/15/2011 10:...	860	00081-20110915-01	PANEL-F-57...	1085865667	1 DUO;	
	9/15/2011 10:...	865	00081-20110915-01	PANEL-A-5...	1085865668	1 DUO;	
	9/15/2011 10:...	870	00081-20110915-01	PANEL-B-5...	1085865669	1 DUO;	
	9/15/2011 10:...	875	00081-20110915-01	PANEL-D-5...	1085865670	1 DUO;	
	9/15/2011 10:...	880	00081-20110915-01	PANEL-F-57...	1085865620	1 DUO;	
	9/15/2011 10:...	885	00081-20110915-01	PANEL-A-5...	1085865569	1 DUO;	
	9/15/2011 10:...	890	00081-20110915-01	PANEL-B-5...	1085865570	1 DUO;	
	9/15/2011 10:...	895	00081-20110915-01	PANEL-D-5...	1085865571	1 DUO;	
	9/15/2011 6:5...	901	00081-20110915-01	011159626...	1085863001	1 DUO;	

Search By WorklistID

- 1) Click on the 'Search By WorklistID' button and enter the work list identification number in the 'Work List ID' text field.
- 2) Click on the 'Search' button to generate the test order list that is associated to this WorklistID.

Search

Selections

Test Order Look Up

☐ Search By Date Range
 ☒ Search By WorkListID

Start
 End
 Work List ID

Drag a column here to group by this column.

Graph	Creation Time	Test Or...	Work List ID	Sample ID	MTU ID	Posi... ▲	Flags
	9/14/2011 10:...	156	00081-20110...	PANEL-A-5...	1085867312	1 DUO;	
	9/14/2011 10:...	161	00081-20110...	PANEL-B-5...	1085867314	1 DUO;	
	9/14/2011 10:...	166	00081-20110...	PANEL-D-5...	1085867367	1 DUO;	
	9/14/2011 10:...	171	00081-20110...	PANEL-F-57...	1085866194	1 DUO;	
	9/14/2011 10:...	176	00081-20110...	PANEL-A-5...	1085866195	1 DUO;	
	9/14/2011 10:...	181	00081-20110...	PANEL-B-5...	1085866196	1 DUO;	
	9/14/2011 10:...	186	00081-20110...	PANEL-D-5...	1085866197	1 DUO;	
	9/14/2011 10:...	191	00081-20110...	PANEL-E-56...	1085866198	1 DUO;	
	9/14/2011 10:...	196	00081-20110...	PANEL-A-5...	1085866199	1 DUO;	

The 'Reset Dates' button adjusts the 'Start Date' and 'End Date' to the past seven days, ending with the present date; it also clears the 'Work List ID' text field.

The 'Explode Current Graph' button enlarges the most recent selected graph.

The 'Print Current Graph' button prints the most recent selected graph.

Test Order Look Up – Module Components

Column Headers

Graph – selects an entire test order row.

Creation Time – local time and date that the test order was created.

Test Order ID – test order identification number.

Work List ID – work list identification number.

Sample ID – sample identification number.

MTU ID – Multi-Tube Unit identification number.

Position In MTU – Multi-Tube Unit position number; 1 MTU = 5 disposable reaction tubes.

Selections

The ‘*Selections*’ tab displays the test orders that have been selected to generate a graph(s). It is located above the ‘*Test Order Look Up*’ screen, next to its ‘*Search*’ tab.

The ‘*Clear*’ button removes the list of test orders that were selected on the ‘*Test Order Look Up*’ in the ‘*Search*’ tab.

Search

Selections

Selected Entries to Graph

Clear

Drag a column here to group by this column.

Del	Creation Time ▼	Test O...	Work...	Sam... ▲	MTU ID	Positi...	Flags
	9/15/2011 8:42:56 PM	760	00081-...	PANEL-A...	1085863978	1	DUO;
	9/15/2011 7:57:56 PM	715	00081-...	PANEL-A...	1085863969	1	DUO;
	9/15/2011 7:22:56 PM	680	00081-...	PANEL-B...	1085863009	1	DUO;
	9/14/2011 10:47:22 PM	206	00081-...	PANEL-C...	1085866201	1	DUO;
	9/14/2011 10:46:57 PM	387	00081-...	0111589...	1085867310	1	DUO;

Test Order Column Headers

Test Order Look Up columns can be organized by sorting, conditional formatting, and grouping by right-clicking the column header and selecting the listed functionality.

Search

Selections

Test Order Look Up

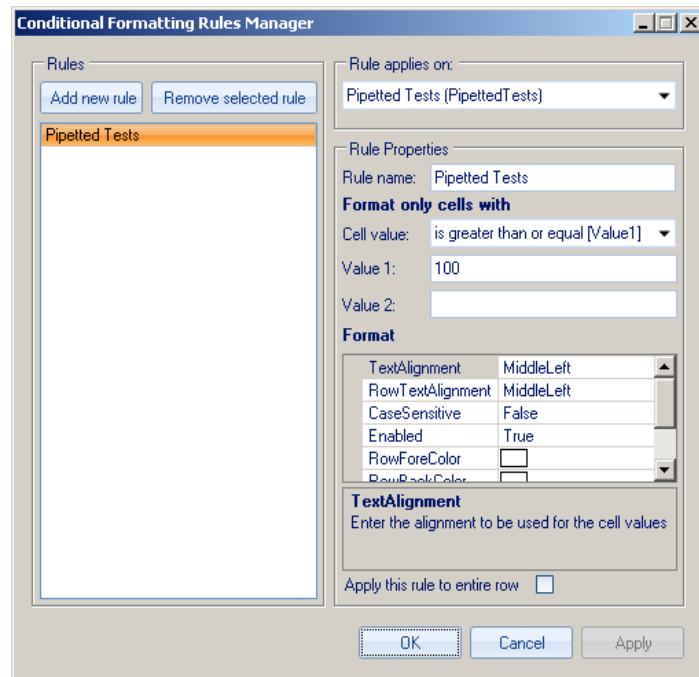
☒ Search By Date Range
 ☐ Search By WorkListID

Start
 End
 Work List ID

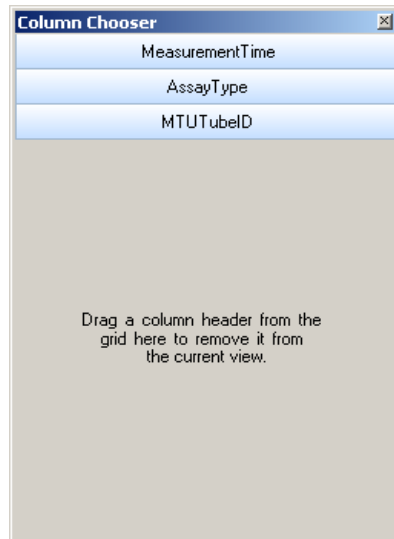
Drag a column here to group by this column.

Graph	Creation Time	Test Order ID	Work List ID	Sample ID	MTU ID
	9/15/2011 6:52:56 PM	902		L258923915011306395	1085863001
	9/15/2011 6:52:56 PM	901		L159626115061301132	1085863001
	9/15/2011 10:57:57 PM	899		NEL-E-567424	1085865571
	9/15/2011 10:57:57 PM	898		NEL-E-567424	1085865571
	9/15/2011 10:57:56 PM	897		NEL-D-568069	1085865571
	9/15/2011 10:57:56 PM	896		NEL-D-568069	1085865571
	9/15/2011 10:57:56 PM	895		NEL-D-568069	1085865571
	9/15/2011 10:52:56 PM	894		NEL-C-571575	1085865570
	9/15/2011 10:52:56 PM	893		NEL-C-571575	1085865570

- Columns can be sorted in ascending or descending order by right-clicking on the column header and selecting 'Sort Ascending' or 'Sort Descending.'
- Column sorting can be reversed by right-clicking on the column header and selecting 'Clear Sorting.'
- Conditional Formatting can be done by right-clicking on the column header and selecting 'Conditional Formatting.' This feature assists with the creation of rules for customized formatting. The following can be controlled: text alignment, row alignment, case sensitive, row color, and cell color.



- 4) Columns can be grouped by an existing column by right-clicking on the column header and selecting 'Group by this column.' Grouping can be used to consolidate non-numeric data values into more general group values; these group values can be mapped into a new column within the data set.
- 5) Columns can be removed from the current view by right-clicking on the column header and selecting 'Column Chooser.' Drag a column header into the gray area in the 'Column Chooser' dialog box to remove the column.
- 6) Columns can be hidden from the Event Logs screen by right-clicking on the column header and selecting 'Hide Column.' To unhide the column, right-click on any column header and select 'Column Chooser,' then drag the hidden column header back out to the 'Test Order Look Up' screen.



- 7) Columns can be 'pinned' so that they remain at the ends of the screen (left or right side); this keeps the column headings visible as the data bar is scrolled up and down. To pin the column, right-click on the column header and select '*Pin Column;*' to reverse this, right-click on the same column header and select '*Unpin Column.*'
- 8) Columns do not automatically accommodate to fit the information they contain; however, by right-clicking and selecting '*Best Fit,*' the column automatically expands and contracts to fit the data enclosed.

Generating a Graph

- 1) In the '*Graphing*' module and on the '*Test Order Look Up*' screen, select at least one test order by clicking on the test order '*Graph*' column; this highlights the entire test order row. Select as many test orders as needed.

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Search Selections

Test Order Look Up

☒ Search By Date Range ☐ Search By WorkListID

Start: Mon, Mar 22, 2010 00:00:00
End: Mon, Oct 24, 2011 00:00:00

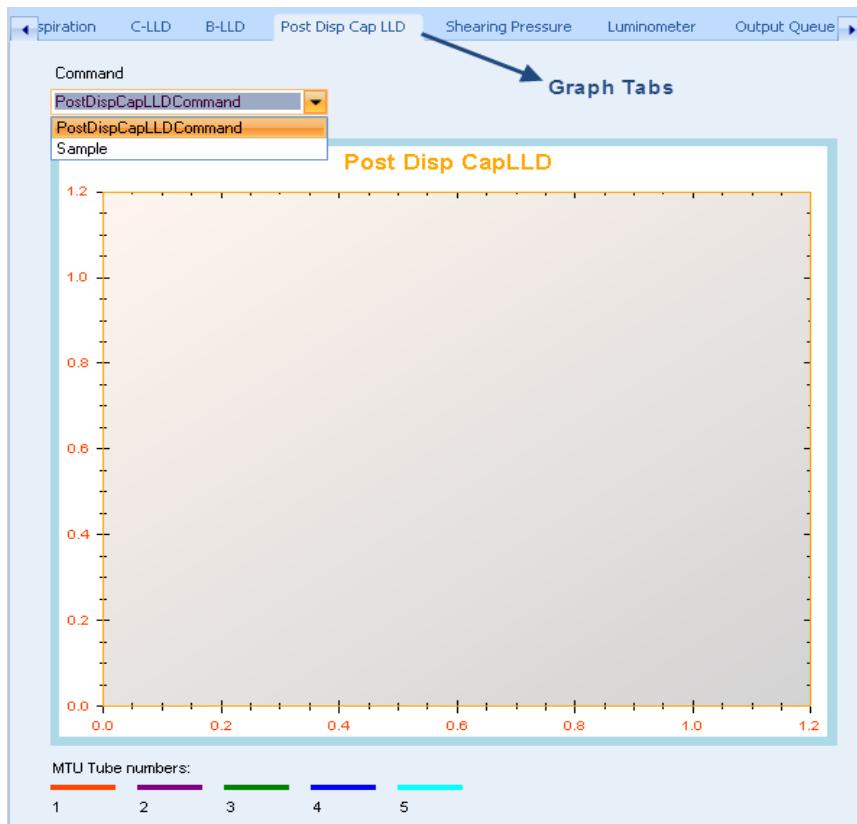
Work List ID:

Drag a column here to group by this column.

Graph	Creation Time	Test Order ID	Work List ID	Sample ID	MTU ID	Positio...	Flags
	9/9/2011 7:35:...	3	00081-2011...	PANEL-A-5...	1085866632	5 DUO;	
	9/9/2011 7:40:...	8	00081-2011...	PANEL-B-5...	1085866634	5 DUO;	
	9/9/2011 7:45:...	13	00081-2011...	PANEL-D-5...	1085866636	5 DUO;	
	9/9/2011 7:50:...	18	00081-2011...	PANEL-E-5...	1085866637	5 DUO;	
	9/9/2011 7:55:...	23	00081-2011...	PANEL-A-5...	1085866638	5 DUO;	

Selecting a test order to graph: click on the 'Graph' column.

- 2) On the right side of the 'Graphing' screen, select the desired graph by clicking on the graph's tab.



- 3) Go to the graph's 'Command' drop-down menu and select the desired parameter. Once the parameter is selected, the graph will automatically plot.

How to Graph an Aspiration Pressure Curve

- 1) In the 'Graphing' module and on the 'Test Order Look Up' screen, select a sample by clicking on the 'Graph' column; this highlights the sample's order row.
- 2) On the right side of the 'Graphing' screen, select the 'Aspiration' tab.
- 3) To plot a graph, select a 'Command' from the drop-down menu.

REPORTS

Reports are used as snapshots of flags and errors that occur in the PANTHER system. They display flags and error codes by module, severity, and frequency of occurrence for a given period of time. They also contain tables of process control data for several modules.

HEALTH REPORTS

Health Reports are made up of a series of sub reports with content determined by the date/ time range specified for the report. Each health report type contains a specific set of sub reports. Health Reports consist of the following grouped sections:

- A report header which contains information that identifies the site, report run date and time, and report title.
- A report body that contains the sub reports that make up the overall report.
- Page footers that contain the report date range, page numbers, and instrument serial number.

The following health reports can be run using the Dashboard application:

PANTHER Health Report

A health report summarizing test order activity, flagged test orders, events, and diagnostic information for a specified date range. This report is created if greater than 32 hours has elapsed from the previously generated report.

The PANTHER Health Report contains the following sub reports:

- Instrument Configuration
- Assay Test Order Count
- Total Test Orders with Flags
- Test Orders with Single Flags
- Test Orders with Multiple Flags
- Flags by Module
- Notable Events
- Module To Severity Level Break Down
- Event Numbers for Modules, Submodules
- Threshold Table – Lumo and Output Queue
- MagWash Table

One-day Health Report

A health report summarizing test order activity, flagged test orders, events, and detailed diagnostic information for a specified 32 hour period. This report is created if less than 32 hours has elapsed from the previously generated report.

The One-day Health Report contains the following sub reports:

- Instrument Configuration
- System Statistics
- Assay Test Order Count
- Rate of Invalids
- Total Test Orders with Flags

- Test Orders with Single Flags
- Test Orders with Multiple Flags
- Flags by Module
- Notable Events
- Module To Severity Level Break Down
- Event Numbers for Modules, Submodules
- Threshold Table – Lumo and Output Queue
- MagWash Table
- Flags
- Sensor Readings

System Health Report

A health report summarizing test order activity and flagged test orders for a specified data range.

The PANTHER Health Report contains the following sub reports:

- Instrument Configuration
- System Statistics
- Assay Test Order Count
- Rate of Invalids
- Total Test Orders with Flags
- Test Orders with Single Flags
- Test Orders with Multiple Flags

Manufacturing Health Report

A health report summarizing test order activity, flagged test orders, events, and detailed diagnostic information for a date/ time range derived from selected work lists. It is intended for use by Manufacturing for diagnosing saline test runs.

The Manufacturing Health Report contains the following sub reports:

- Instrument Configuration
- System Statistics
- Assay Test Order Count
- Rate of Invalids
- Total Test Orders with Flags
- Test Orders with Single Flags
- Test Orders with Multiple Flags
- Flags by Module
- Notable Events
- Module To Severity Level Break Down
- Event Numbers for Modules, Submodules
- Threshold Table – Lumo and Output Queue

- MagWash Table
- Flags
- Sensor Readings

The screenshot displays the PANTHER software interface. On the left, a 'Select Report' dropdown menu is open, showing a list of reports including 'Panther Health Report', 'One-day Health Report', 'System Health Report', and 'Manufacturing Health Report'. Below the dropdown is a table with columns for 'Start', 'End', and 'ID'. The 'Manufacturing Health Report' is selected. A 'Preview' button is located below the table. On the right, the 'Manufacturing Health Report' is displayed. It includes the PANTHER logo, the date 'Thursday, October 20, 2011 6:33:43 PM', and the title 'Manufacturing Health Report'. Below the title is a section titled 'Instrument Configuration' which contains the following information:

Instrument Configuration		
Software Version(s)	9/15/2011 4.2.0.11	Current
	9/8/2011 4.2.0.9	
Firmware Version(s):	10/20/2011	Problem processing Firmware version information
Installed Assays(s):	9/15/2011 HIV/HCV/HBV_1.26.1	
	9/15/2011 dHIV_1.25.1	
	9/15/2011 dHCV_1.24.1	
	9/15/2011 dHBV_1.24.1	
	9/15/2011 HPV_1.48.1	
	9/15/2011 HPV_1.50.1	
	9/15/2011 GC_1.39.1	
	9/15/2011 GC_1.39.1	
	9/15/2011 CT_1.36.1	
	9/15/2011 CT_1.39.1	
	9/15/2011 CT/GC_1.33.1	
	9/8/2011 HIV/HCV/HBV_1.24.1	
	9/8/2011 SysCheck_1.10	
	9/8/2011 Flashcheck_1.15	
	9/8/2011 dHIV_1.23.1	
	9/8/2011 dHCV_1.22.1	
	9/8/2011 dHBV_1.22.1	
	9/8/2011 HPV_1.48.4	
	9/8/2011 HPV_1.50.4	
	9/8/2011 GC_1.39.4	
	9/8/2011 GC_1.39.4	
	9/8/2011 CT_1.36.4	
	9/8/2011 CT_1.39.4	
	9/8/2011 CT/GC_1.33.4	
	9/8/2011 CT/GC_1.40.1	

How to Run a Health Report

- 1) Select Report – choose a report from this drop-down list.
 - 2) Report Date – choose the date to run the report for.
 - 3) Preview – click on the 'Preview' button (on the left side of the screen) to generate the report.
- The report is now displayed on the right side of the screen.

The following buttons can be used during or after running a report:

Stop  – stops the report from generating.

Refresh  – starts the report generation from scratch.

First Page  – jumps to the first page of the report.

Previous Page  – jumps to the previous page of the report.

Page field – displays the current page the report is on; can edit the field with a different page number.

Next Page  – jumps to the next page of the report.

Last Page  – jumps to the last page of the report.

Print  – prints the report.

Page Setup  – select the paper size, orientation, and margins.

Export  – exports to Acrobat (PDF) file.

Zoom – enlarges or reduces the display view on the right screen.

Health Report Sections – Defined

Software Version(s)

System History with the software installation dates and software version numbers.

Firmware Version(s)

System History with the firmware installation dates and firmware version numbers.

Installed Assays(s)

The Installed Assays(s) version numbers and their installation dates.

System Statistics

Statistics of the Instrument: Uptime, Number of Restarts and Number of Fatal Errors that occurred since the previously generated report.

Assay Test Order Count

Assay types that ran on the instrument and the total number of assays executed over the duration of the report.

Rate of Invalids

Statistics of the Instrument: Number of Tests, Number of Invalids and Rate of Invalids that occurred since the previously generated report.

Total Test Orders with Flags

The number of test orders attempted on the system with the percentage of samples flagged during the interval of the report generated.

Test Orders with Single Flags

The number of test orders released with a single flag and the corresponding percentage.

Test Orders with Multiple Flags

Number of times this specific confirmation of flags occurred over the time interval of the report generated.

Flags By Module

The number of flags that occurred per instrument module.

Notable Events

A Subset of events that occurs during the instrument activity and message log events filtered for instrument critical events. I.E. Prime, Hardware Failures, Non Fatal Hardware Events and Issues.

Event Numbers for Modules per Sub Module

- Message – **Green text**
- Warning – **Orange text**
- Critical – **Red text**
- Fatal – **Red background with black text**

Health Report Summary

	One-Day Health Report	System Health Report	PANTHER (System) Health Report
For Technical Support Use	X	N/A	X
For Field Service Use	X	N/A	X
For Marketing Use	X	X	N/A
Software Version(s)	X	X	X
Firmware Versions(s)	X	X	X
Installed Assay(s)	X	X	X
System Statistics	X	X	X
Assay Test Order Count	X	X	X
Rate of Invalids	X	X	X

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PANTHER Dashboard User Guide

Proprietary Information

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Total Test Orders with Flags	X	X	X
Test Orders with Single Flags	X	X	X
Test Orders with Multiple Flags	X	X	X
Flags by Module	X	N/A	X
Notable Events	X	N/A	X
Module to Severity Level Breakdown	X	N/A	X
Event Numbers for Modules, SubModules	X	N/A	X
Threshold Table – Lumo and Output Queue	X	N/A	X
MagWash Table	X	N/A	X
Flags	X	N/A	N/A
Sensor Readings	X	N/A	N/A

RLU EVALUATIONS

RLU Evaluations are conducted in following four modules:

- Flash-Check
- System-Check
- HPV-PQ
- AC2-PQ

FLASHCHECK

The *Flashcheck* Reagent is intended for use in performing Quality Control verification of the Auto Detect fluid pumps and injectors in the PANTHER Luminometers as part of new instrument Manufacturing and existing instrument Service/Maintenance procedures. The PANTHER Luminometers are self-contained microprocessor based instruments that measure Relative Light Unit (RLU) values and report raw data for further analysis. The software in these instruments allow protocol programming, perform quality control calibration, acquire RLU values and monitor various sensors to warn the user of possible invalid conditions. The instruments contain pumps and injectors that dispense detection reagents (Auto Detect Reagents) that create chemiluminescent reactions. The *Flashcheck* Reagent is a chemiluminescent oligonucleotide complex dissolved in an acidic buffered detergent-based solution. Flash Check shall be developed as a system check to measure the performance of High Count and Chiron Luminometers to ensure that assay results are consistent within an instrument and from instrument to instrument. *Flashcheck* is a chemiluminescent formulation that generates a signal on PANTHER instruments by sequential mixing with a hydrogen peroxide reagent (Auto Detect 1) and a sodium hydroxide reagent (Auto Detect 2).

The *Flash Check* software module is responsible for displaying the results of a *Flashcheck* Assay reagent test(s). The results of these tests consist of measuring luminescence from 4 PANTHER MTU (Multi Tube Unit) x 5 replicates totaling 20 replicate samples. Each replicate luminescence measurement consists of 200, 10–msec *RLU (Relative Light Unit)* sample measurements from the Luminometers found in each of the 4 MTUs of the PANTHER RT Incubator. The 200 sample measurements are then converted into fifty intervals containing 40-msec RLU values by summing every four cells and storing the values. From there, average and variant statistical values are calculated to determine acceptance criteria. The system shall also indicate whether there are outliers or no outliers.

The screenshot displays the PANTHER software interface. The top navigation bar includes tabs for Event Logs, Flags by Worklist, Graphing, Health Reports, **Flash Check**, Sys Check, Quantitative Graphing, AC2 PQ, HPV PQ, LIS Monitor, and PRO 360 Connectivity Tester.

View Previous Reports: A dropdown menu shows a list of reports: FlashCheckReport_Jun11_2012_033802PM, FlashCheckReport_Jun11_2012_033943PM, FlashCheckReport_Jun11_2012_040035PM, and FlashCheckReport_Jun11_2012_040051PM (highlighted). Below this, a 'Notes' field contains 'No notes entered by the user'.

Operator Information: Fields for Operator Name, Today's Date (6/11/2012 4:12:21 PM), Instrument ID # (00176), Flashcheck Reagent Lot # (3232), FlashCheck Reagent Expiration Date (Tuesday, November 08, 2011), Selection Reagent Lot # (9898), and Selection Reagent Expiration Date (Tuesday, November 08, 2011) are visible.

Curve Data: A table shows the following data:

Start	End	Runs	Message
11/8/2011 10:54:51 PM	11/8/2011 11:19:10 PM	20	

Buttons for 'Refresh' and 'Perform Calculation' are at the bottom of the sidebar.

FLASHCHECK REPORT: The main report area displays the PANTHER logo and a title 'FLASHCHECK REPORT'. Under 'User Inputs', it shows the same operator information as the sidebar. At the bottom right, it indicates 'Page: 1 of 4'.

Flash Check – Module Components

View Previous Reports – select any of the previous reports ran and view them in PDF format.

User Inputs:

Notes – enter notes related to the Quality Control verification.

Operator Name – enter the name of the instrument user.

Today's Date – local date and time is automatically displayed for tracking.

Instrument ID # – enter the instrument's identification number.

Flashcheck Reagent Lot # – enter the reagent lot number, this is a required field.

FlashCheck Reagent Expiration Date – select the reagent expiration date.

Selection Reagent Lot # – enter the reagent's lot number, this is a required field.

Selection Reagent Expiration Date – select the reagent's expiration date.

Curve Data Column Headings:

Start – run start local date and time.

End – run end local date and time.

Runs – the total amount of runs.

Message – diagnostic message.

Buttons (Left Screen):

Refresh – updates the current *Curve Data* grid view.

Perform Calculation – runs FlashCheck report.

Buttons (Right Screen):

Stop  – stops the report from generating.

Refresh  – starts the report generation from scratch.

First Page  – jumps to the first page of the report.

Previous Page  – jumps to the previous page of the report.

Page field – displays the current page the report is on; can edit the field with a different page number.

Next Page  – jumps to the next page of the report.

Last Page  – jumps to the last page of the report.

Print  – prints the report.

Page Setup  – select the paper size, orientation, and margins.

Export  – exports to Acrobat (PDF) file.

Zoom – enlarges or reduces the display view on the right side screen.

SYS CHECK

The *System Check* software module is responsible for verifying that PANTHER luminometer is in compliance with performance validity and acceptance criteria defined in the Software Requirements Specification document. The software will run several statistical evaluations with the Syscheck Assay of 12 replicates. The result report displays the test demographics along with the user input data. It also includes a calculated calibration factor with a listing of all the test-orders Raw RLU and Calibrated RLU values. The last page of the report displays additional calculated statistical results such as “*Average Calibrated RLU*”, “*Standard Deviation*”, “*Covariance*”, “*Dark Counts*” among others. The software module follows a similar processing methodology as the FlashCheck module.

Event Logs

Flags by Worklist

Graphing

Health Reports

Flash Check

Sys Check

Quantitative Graphing

AC2 PQ

HPV PQ

LIS Monitor

PRO 360 Connectivity Tester

View Previous Reports:

Select any of the previous reports

SysCheckReport_Oct18_2011_033246PM.f

SysCheckReport_Oct20_2011_063815PM.f

Notes:

No notes entered by user...

Operator Name:

Amanda Smith

Today's Date:

10/20/2011 6:37:46 PM

Instrument ID:

00081

Set RLU from the SysCheck Tube:

2

Data

Start	End	WorkListID
9/14/2011 11:41:25 PM	9/14/2011 11:58:46 PM	

Refresh Grid

Perform Calculation

Problem retrieving customer information

PANTHER

SYSHECK REPORT

User Inputs

Notes:

No notes entered by user...

Operator Name:

Amanda Smith

Today's Date:

10/20/2011 6:38:15 PM

Run# :

Run Start Date :

9/14/2011 11:41:25 PM

Run End Date :

9/14/2011 11:58:46 PM

Instrument ID:

00081

Set RLU from the SysCheck Tube:

2

Previous Calibration Factor:

0.193

Raw Data

Tube Number	Raw RLU	Calibrated RLU
0	536608	103565.3
1	531317	102544.2
2	542924	104784.3
3	545601	105301
4	549424	106038.8
5	547601	105687
6	542905	104780.7
7	539129	104051.9
8	534396	103138.4
9	545002	105185.4
10	548152	105793.3
11	545679	105316

Average Calibrated RLU:

104682

Average value within +/- 15% of Syscheck Expected RLU:

False

Results

Page: 1 of 2

Sys Check – Module Components

View Previous Reports – select any of the previous reports ran and view them in PDF format.

User Inputs:

Notes – enter notes related to the Quality Control verification.

Operator Name – enter the name of the instrument user.

Today's Date – local date and time is automatically displayed for tracking.

Instrument ID # – enter the instrument's identification number.

Set RLU from the SysCheck Tube – enter the RLU number, this field is required.

Data Column Headings:

Start – run start local date and time.

End – run end local date and time.

WorklistID – work list identification number.

Buttons (Left Screen):

Refresh Grid – updates the current *Data* grid view.

Perform Calculation – runs FlashCheck report.

Buttons (Right Screen):

Stop  – stops the report from generating.

Refresh  – starts the report generation from scratch.

First Page  – jumps to the first page of the report.

Previous Page  – jumps to the previous page of the report.

Page field – displays the current page the report is on; can edit the field with a different page number.

Next Page  – jumps to the next page of the report.

Last Page  – jumps to the last page of the report.

Print  – prints the report.

Page Setup  – select the paper size, orientation, and margins.

Export  – exports to Acrobat (PDF) file.

Zoom  whole page – enlarges or reduces the display view on the right side screen.

QUANTITATIVE GRAPHING

The *Quantitative Analysis* software module provides the technician a view into the general pressure and/or command curve data over a designated range of time. This curve data is similar to the curve data reported by the *Graphing* module however; the result plots are a composite of curve data over the designated time range. The software retrieves the curve data from the database in incremental requests and provides a continuous plotted update as the data is retrieved. The plotting report controls automatic scaling of the output as new data arrives until the final data retrieval/update is completed for that user's designated time range. Once the user enters a specific WorklistID or selectable date range, then additional filter criteria becomes available. This additional filter criteria selection helps build the final query for the successive curve type requests, some of these selections are listed below. Note that the selections depend on the workflow defined in the executed assay; specifically, the values are presented depending on what the instrument has executed.

- CurveType
 - Aspiration
 - Dispense
 - Shearing
 - BubblePopCLLD1
 - CLLD
 - BubblePopCLLD2
 - PostDispCapLLD
 - MagWash
 - LLD
- Command
 - Amplification
 - Amplification Spitback
 - Enzyme
 - Enzyme Spitback
 - Probe
 - Probe Spitback
 - Sample
 - Select
- Quantitative Fields
 - PMax
 - PRampDown
 - Pend
 - PStatic
 - PMean
 - PRef
 - PRiseDelay

o TDownDelay

Curves Look Up

Please select the search criteria.

☒ Search By WorkListID

Work List ID

or

☐ Search By Date Range

Start End

Curve Type:

Command:

Quantitative Fields:

Quantitative Graphing – Module Components

A graph can be plotted by WorklistID number or by a date range.

Curves Lookup

Search By WorkListID – radio button selects to plot a graph by Work List ID.

Work List ID – field to enter Work List ID number.

Search By Date Range – radio button selects to plot a graph by a date.

Start – run start local date and time.

End – run end locate date and time.

WorkList Id – select a Work List ID from the drop-down menu.

Curve Type – select a curve type for the drop-down menu.

Command – select a command from the drop-down menu.

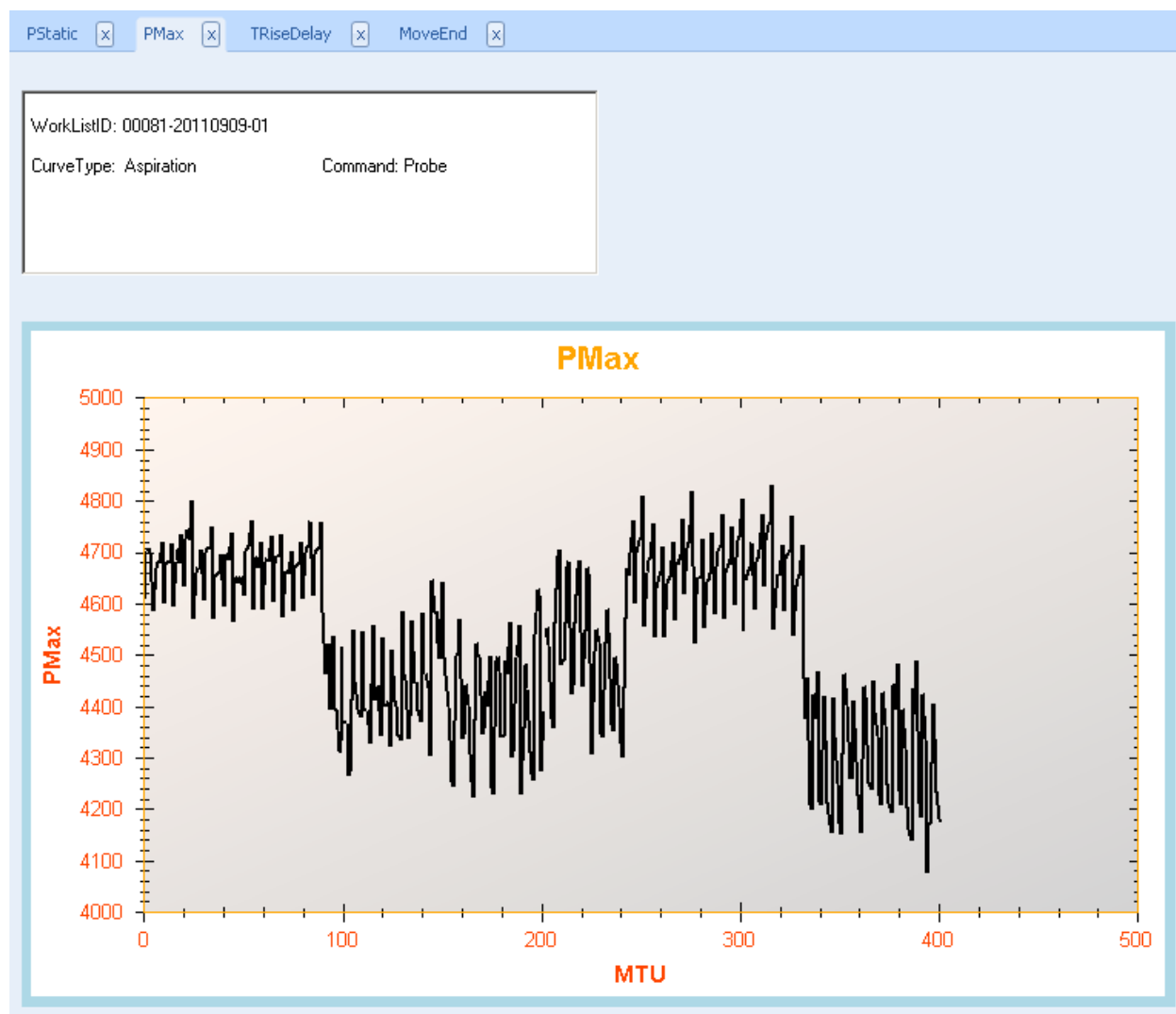
Quantitative Fields – select a quantitative field.

Buttons:

Plot Graph – generate graph.

Explode Current Graph – enlarge or reduces the current graph view.

Print Current Graph – prints the current graph.



AC2 PQ

The AC2 PQ (*Aptima Combo CT/GC Assays Performance Qualification*) module is responsible for validating and verifying the luminescence acceptance criteria for proper PANTHER system operation. The software module processes the luminescence data similarly to the *Sys Check* and *Flash Check* modules in that statistical calculations are performed to verify compliance to the requirement specifications. The AC2 PQ module will retrieve a specific population of panels (Panel A – Panel E) information from the database, based on a user selected run and will calculate and determine the following conditions...

APTIMA COMBO 2 Assay Acceptance Criteria

Target	Expected CT Result	Expected GC Result	RLU Specification	% CV	Allowable Variance
Panel A	Negative	Negative	< 30,000	N/A	0/10
Panel B	Positive	Negative	> 500,000	< 25	0/5
Panel C	Negative	Positive	> 500,000	< 25	0/5
Panel D	Positive	Positive	> 1,000,000	< 25	0/5
Panel E	Positive	Negative	> 100,000	FIO	2/20

Allowable invalid reactions per assay run for APTIMA COMBO 2 Assay:

- 1 – VVFS or VVFT
- 1 – Cap Pierce Failure if caused by a double foil
- Up to 2 Equivocal Panel E results (> 25,000 RLU)
- Verify % CV and Average RLU of Panel B / Panel C ratio between $0.5 < \text{Ratio} < 2.0$ for all valid results.

The output report displays the results of the calculations and whether there is an overall *pass* or *fail* of the complete test based upon the allowable invalid reactions.

AC2 PQ – Module Components

View Previous Reports – select any of the previous reports ran and view them in PDF format.

User Inputs:

Notes – enter notes related to the Quality Control verification.

Operator Name – enter the name of the instrument user.

Today's Date – local date and time is automatically displayed for tracking.

Instrument ID # – enter the instrument's identification number.

Data Column Headings:

Start – run start local date and time.

End – run end local date and time.

WorklistID – work list identification number.

Buttons (Left Screen):

Refresh Grid – updates the current *Data* grid view.

Perform Calculation – runs FlashCheck report.

Buttons (Right Screen):

Stop  – stops the report from generating.

Refresh  – starts the report generation from scratch.

First Page  – jumps to the first page of the report.

Previous Page  – jumps to the previous page of the report.

Page field – displays the current page the report is on; can edit the field with a different page number.

Next Page  – jumps to the next page of the report.

Last Page  – jumps to the last page of the report.

Print  – prints the report.

Page Setup  – select the paper size, orientation, and margins.

Export  – exports to Acrobat (PDF) file.

Zoom – enlarges or reduces the display view on the right side screen.

Report

1

of 4

89%

Problem retrieving customer information

PANTHER

AC2 PQ REPORT

AC2 PQ Results

Operator Name:

Amanda Smith

Today's Date:

10/20/2011 6:53:18 PM

Instrument ID:

00081

Notes:

No notes entered by the user...

Run #:

00081-20110916-01

Start Date:

9/16/2011 7:19:04 PM

End Date:

9/17/2011 2:44:49 AM

Overall PQ Result

PQ Validity:

Fail

Run Summary

Target	Nbr Replicates	Average	% CV	Acceptable
Panel A	10	5300	NA	Yes
Panel B	5	1030800	2.7	Yes
Panel C	5	1151400	1.3	Yes
Panel D	5	2073800	1.3	Yes
Panel E	20	756421	FIO	Yes

Panel B/Panel C Ratio:

0.895257946847316

Number of invalids caused by WFS or WFT:

0

Number of Equivoval Panel E:

0

How many Cap Pierce failures occurred:

0

HPV PQ

The *HPV PQ* module is responsible for validating and verifying the luminescence acceptance criteria for proper PANTHER system operation. The software module processes the luminescence data similarly to the *Sys Check* and *Flash Check* modules in that statistical calculations are performed to verify compliance to the requirement specifications. The *HPV PQ* module will retrieve a specific population of 40 panels (Panel A) information from the database, based on a user selected run and will calculate and determine the following conditions...

APTIMA HPV Assay Acceptance Criteria

Target	Expected Result	N	Expected Result Allowable Variance	RLU Specification	Analyte RLU Allowable Variance
Panel A	Negative	40	0/40	≤ 30,000	1/40

Allowable invalid reactions per assay run for APTIMA HPV Assay:

- 1 – VVFS or VVFT
- 1 – Cap Pierce Failure if caused by a double foil
- Up to 1 Panel A > 30,000 RLU with a negative result

HPV PQ – Module Components

View Previous Reports – select any of the previous reports ran and view them in PDF format.

User Inputs

Notes – enter notes related to the Quality Control verification.

Operator Name – enter the name of the instrument user.

Today's Date – local date and time is automatically displayed for tracking.

Instrument ID # – enter the instrument's identification number.

Data Column Headings:

Start – run start local date and time.

End – run end local date and time.

WorklistID – work list identification number.

Buttons (Left Screen):

Refresh Grid – updates the current *Data* grid view.

Perform Calculation – runs FlashCheck report.

Buttons (Right Screen):

Stop  – stops the report from generating.

Refresh  – starts the report generation from scratch.

First Page  – jumps to the first page of the report.

Previous Page  – jumps to the previous page of the report.

Page field – displays the current page the report is on; can edit the field with a different page number.

Next Page  – jumps to the next page of the report.

Last Page  – jumps to the last page of the report.

Print  – prints the report.

Page Setup  – select the paper size, orientation, and margins.

Export  – exports to Acrobat (PDF) file.

Zoom Whole page  – enlarges or reduces the display view on the right side screen.

LIS MONITOR

The *LIS Log File* monitoring software module provides the technician the ability to view LIS log messages as they are being created by PANTHER from the LIS system. This tool exposes any possible malformed messages that may cause invalid or erroneous data that PANTHER may not be able to handle properly. After turning on the LIS logging features located on the PANTHER system LIS Configuration screen, the LIS Monitor will display all LIS/ ASTM traffic. This allows the FSE to determine if connectivity has been properly configured and allows the FSE to work with the LIS Vendor on the ASTM message formats.

The log file monitor is capable of viewing the two different types of transport message files: RS232 and TCP/IP. Typically one of the two transports will be set up by the technician but never both at the same time. The output display of the monitor shows the last 500 or less messages that were received by PANTHER and will ignore any older messages. The monitor adds new messages received and removes older messages from the 500 message buffer to ensure that the buffer limit is never exceeded.

The top view displays any of the TCP messages while the bottom view displays any of the RS232 messages. A green monitoring button “*File Monitor On*” turns the monitor on or off; it uses the *FileWatcher* event service so that system CPU overhead is minimized when not in use.

Buttons:

Load Logs – LIS log messages being created by PANTHER from the LIS system.

File Monitor Off – turns the monitor on or off.

Clear Logs – removes the displayed logs.

PRO360° CONNECTIVITY TESTER

The *Pro360° Connectivity* module displays information of any RDXM server configuration as network addresses are tested. When this test is run it will produce a status result of each connection by executing a network request to each address that is listed. Overall, this module assists with:

- Viewing the Network Interface Card (NIC) configuration settings.
- Testing TCP/IP address/ port connectivity; it is used to ensure LIS connectivity has been configured properly.
- Determines if the DNS resolution is occurring properly.
- Ensures EFT connectivity exists, this is needed for RSL.
- Performs a basic test to ensure that the PANTHER Instrument NIC has been configured correctly.

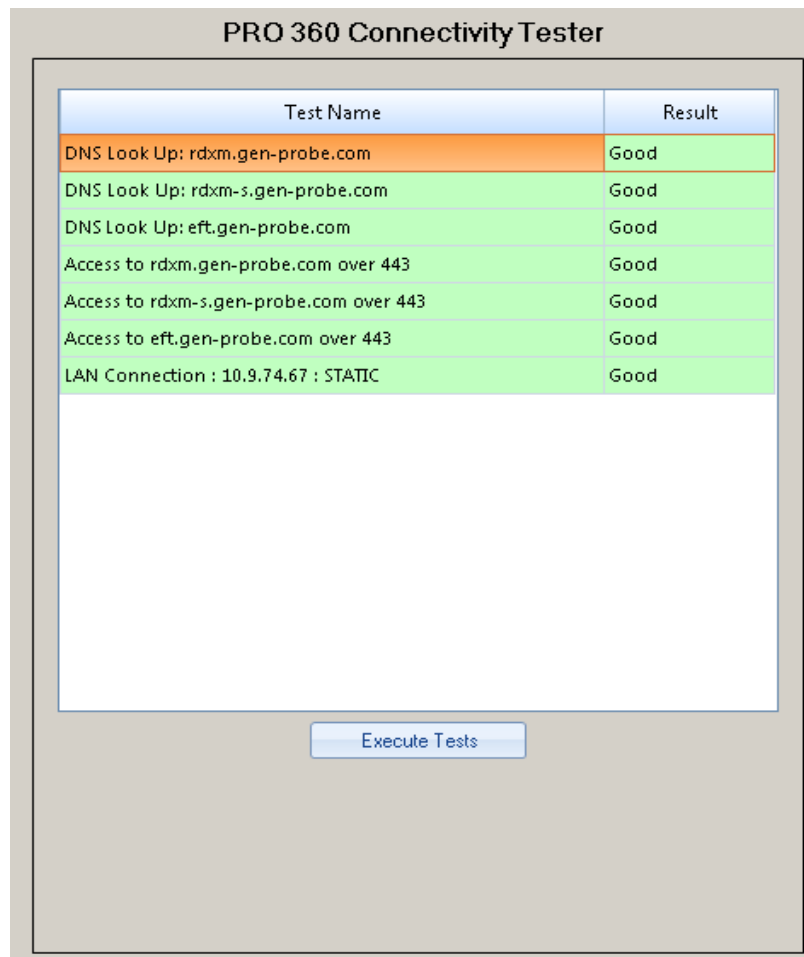
PRO360° Connectivity Tester (*Columns*)

Test Name – displays the network address being tested.

Result – displays network connection result.

Button (Left Screen):

Execute Tests – runs the network test.



NETWORK CONNECTIVITY TESTER

The *Network Connectivity Tester* module is used to determine if a network route exists, if a server is accepting network connections, and optionally if DNS can be resolved. For example, this tester can be

used to verify that the LIS is accessible and accepting connection to the LIS Host. Another use of this tester is to determine if the PANTHER has the ability to access the RDxM servers.

Network Connectivity Tester (Columns)

Test Name – displays the network address being tested.

Result – displays network connection result.

IP or Host Name – enter the network address or a DNS name to test.

TCP/IP Port Number – enter the port number to test.

Network Connectivity Tester

IP or Host Name: TCP/IP Port Number:

Test Name	Result
DNS resolution: rdxm.gen-probe.com	192.168.10.19
rdxm.gen-probe.com over 443	Good

If a DNS name is entered, this tester will first try to resolve the address. Then using this address, this tester will try to establish a network connection to address over the entered port number.

If an IP address is entered, this tester will try to establish a network connection to address over the entered port number.

Button (Right Screen):

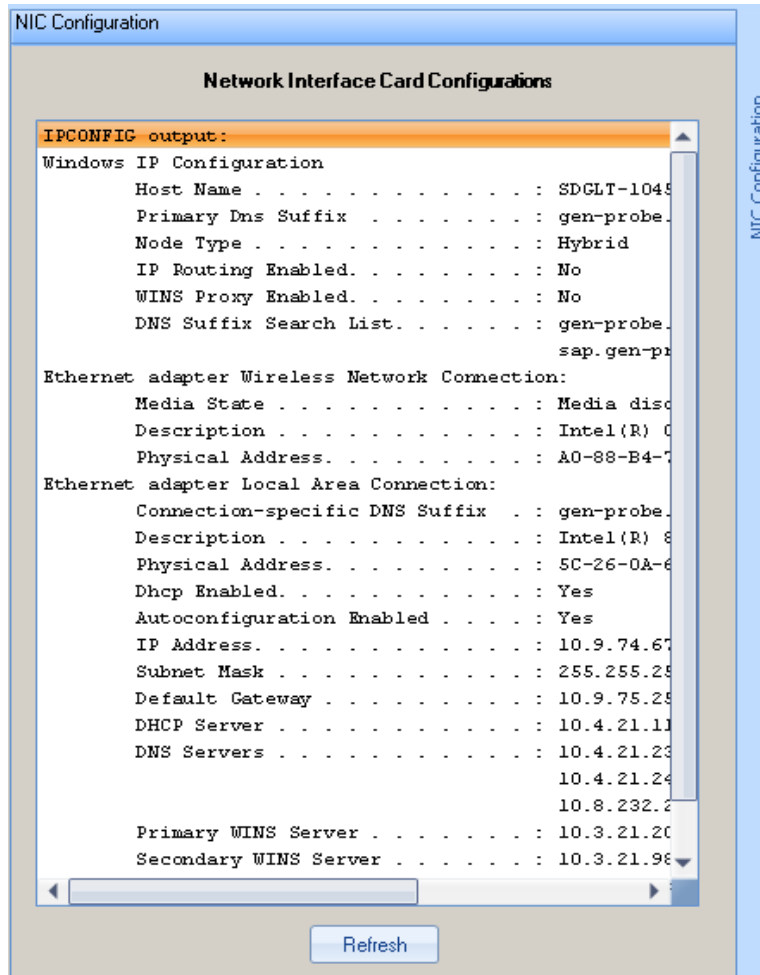
Execute Test – runs the network test.

NIC CONFIGURATION

The *NIC Configuration* component displays the Network Interface Card (NIC) configuration settings.

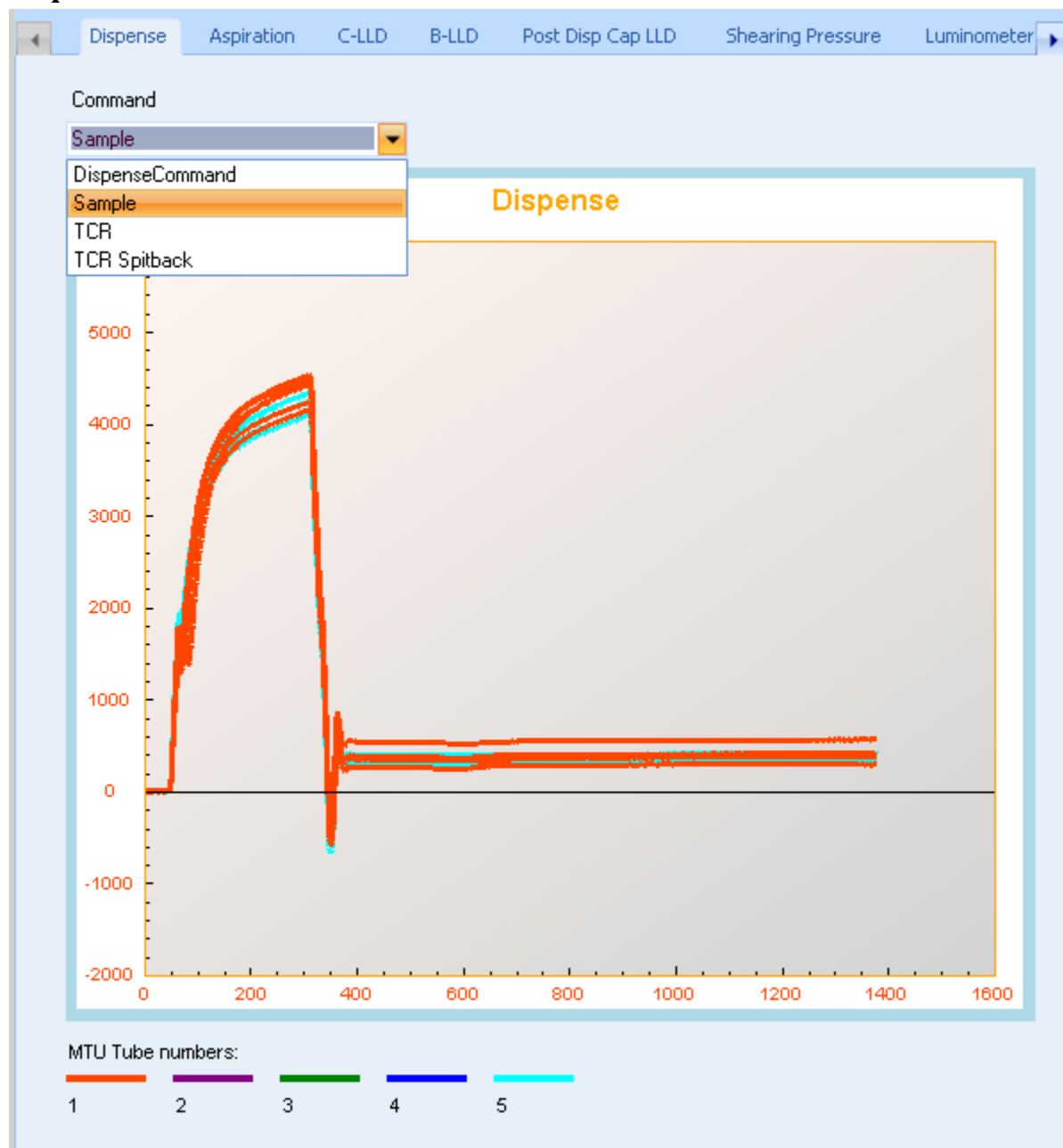
NIC Configuration – This is a tab that is located on the top right side of the screen. Click on the “*NIC Configuration*” tab to view NIC settings.

Refresh – click on this button to update the current Network Interface Card Configurations view.

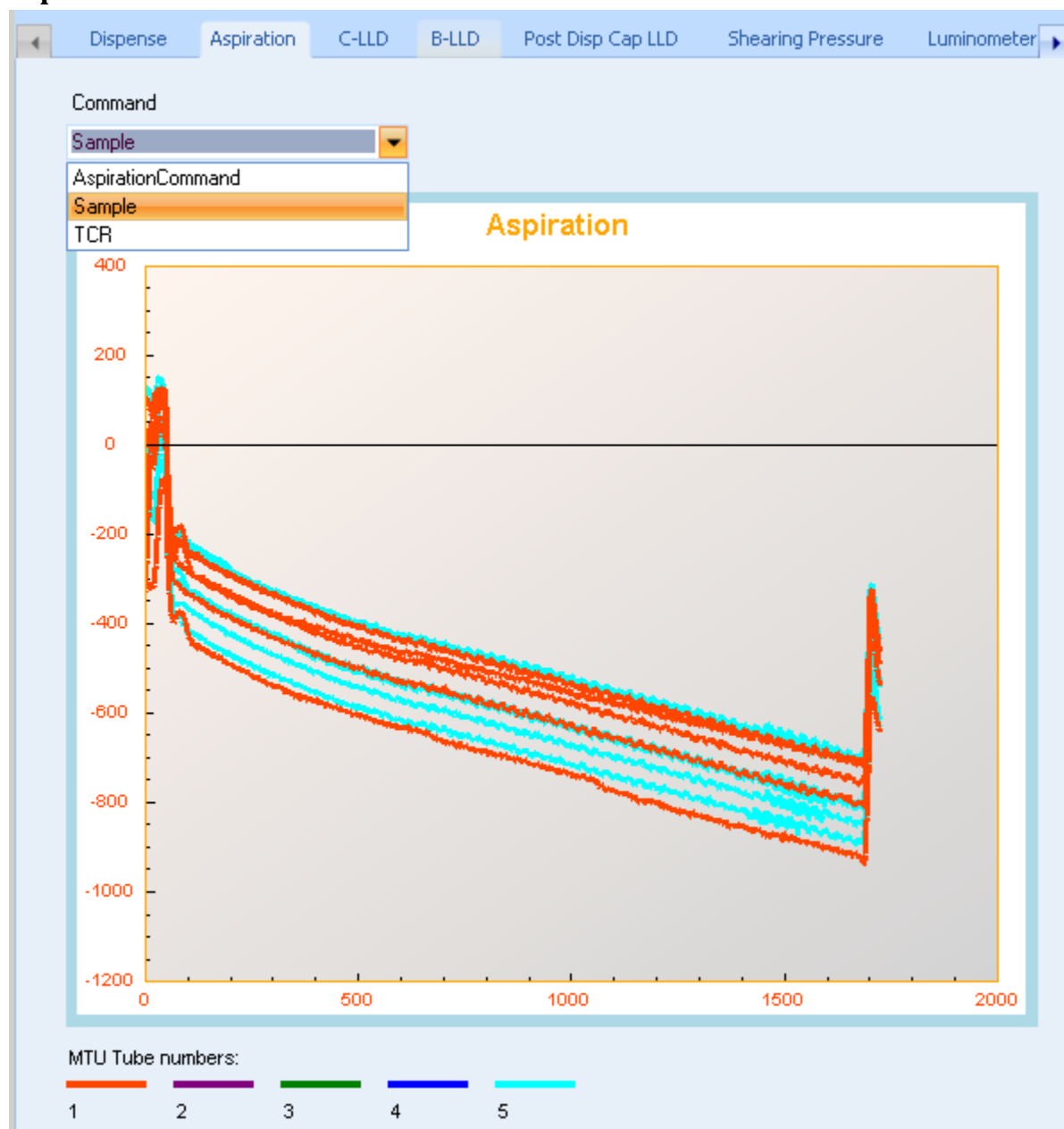


APPENDIX A – TYPES OF GRAPHS

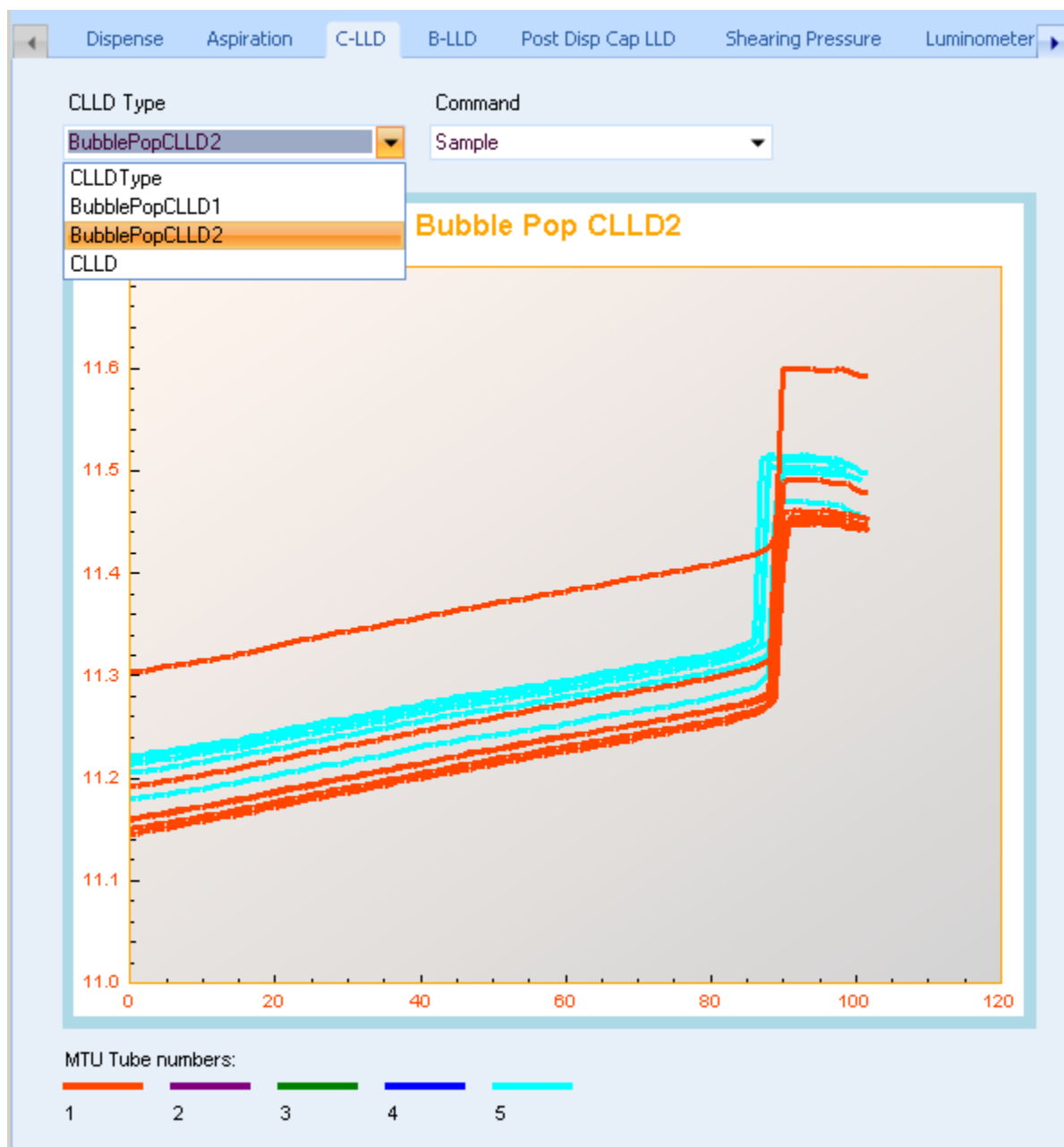
Dispense



Aspiration



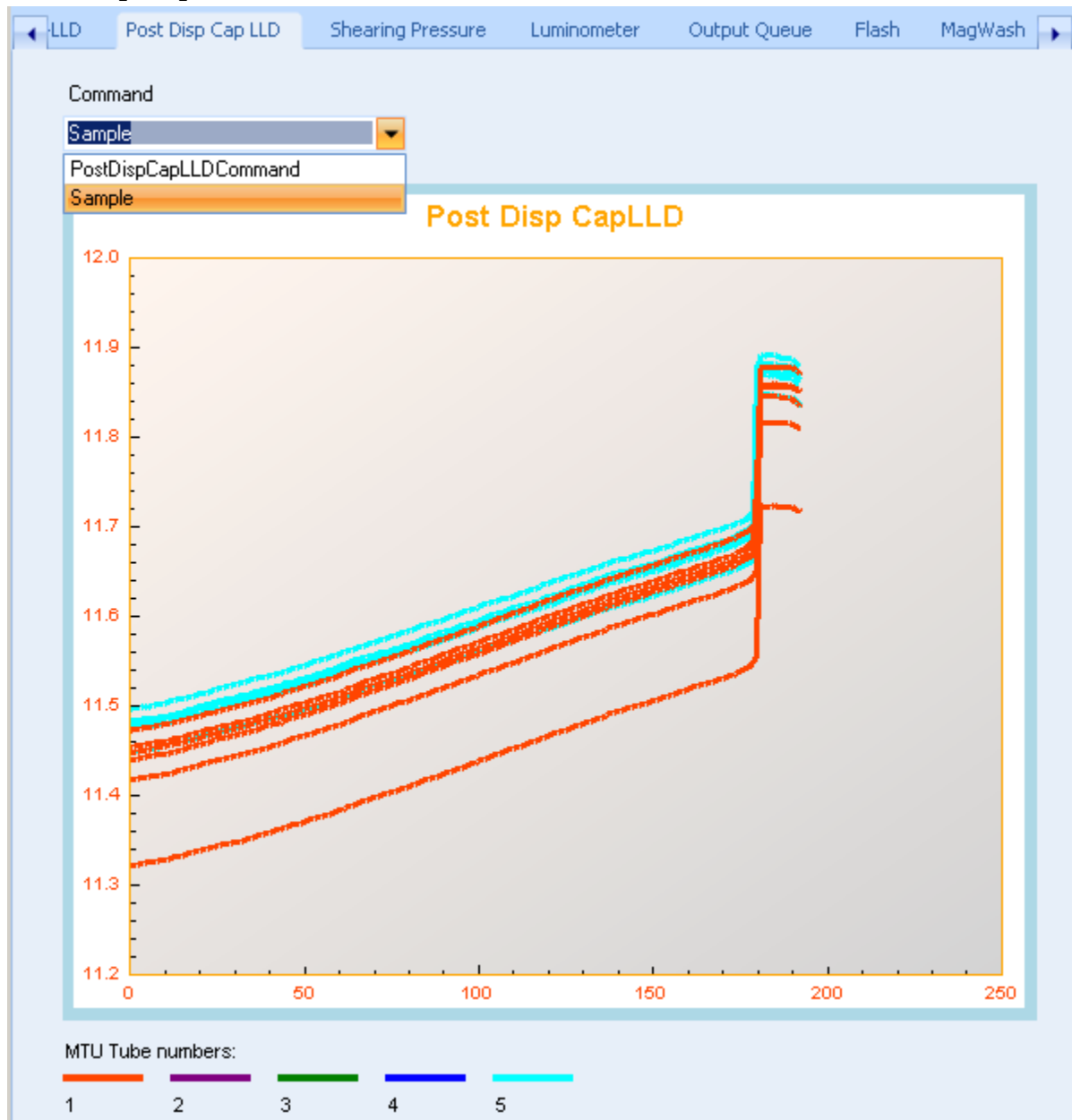
C-LLD



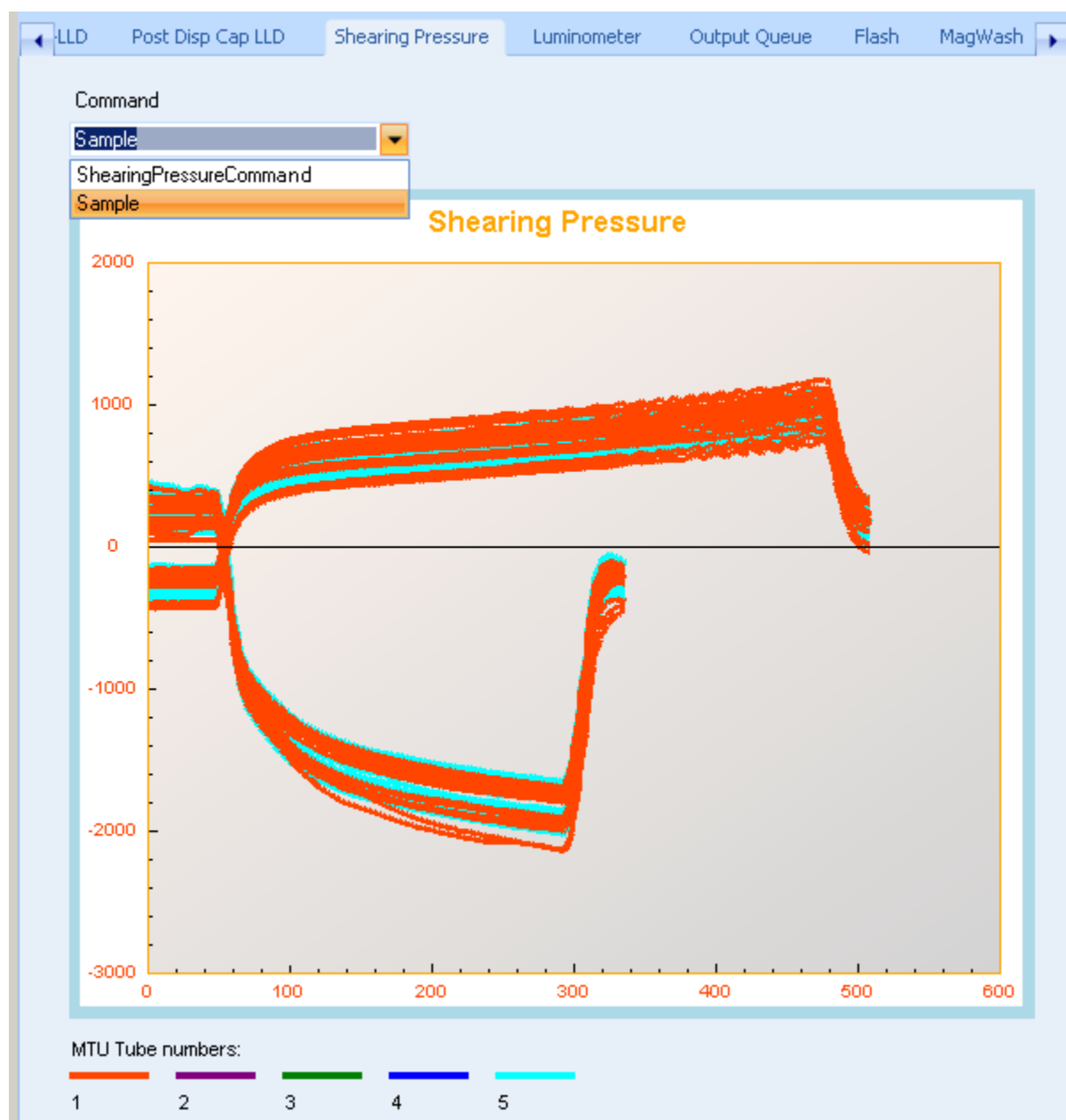
B-LLD

The B-LLD graph is not available because the B-LLD trace logging is disabled on all released builds of the PANTHER software. This setting is omitted due to increased traffic on the CAN Bus; this is particularly the case with the Real-Time Fluorometers that are installed.

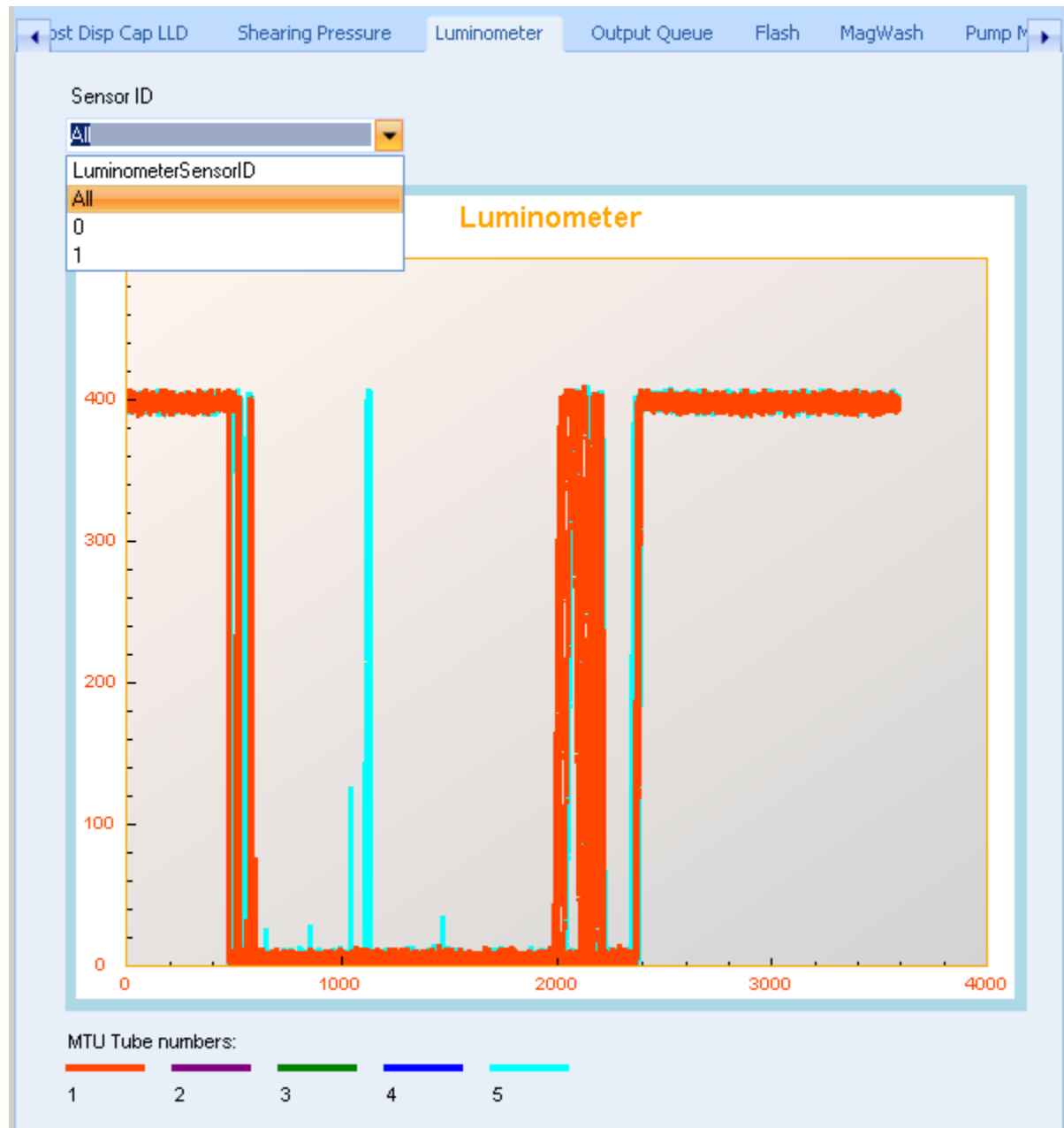
Post Disp Cap LLD



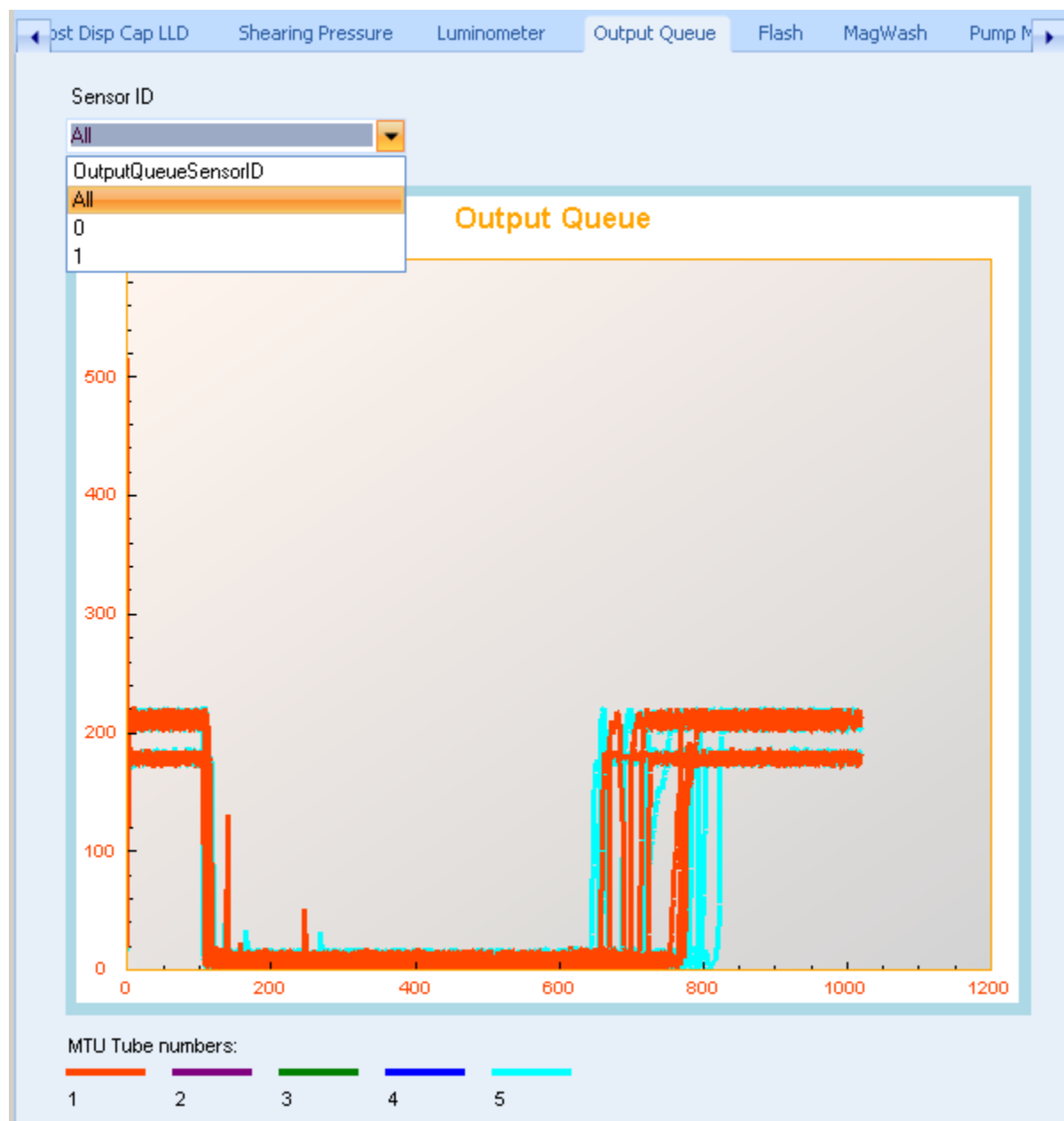
Shearing Pressure



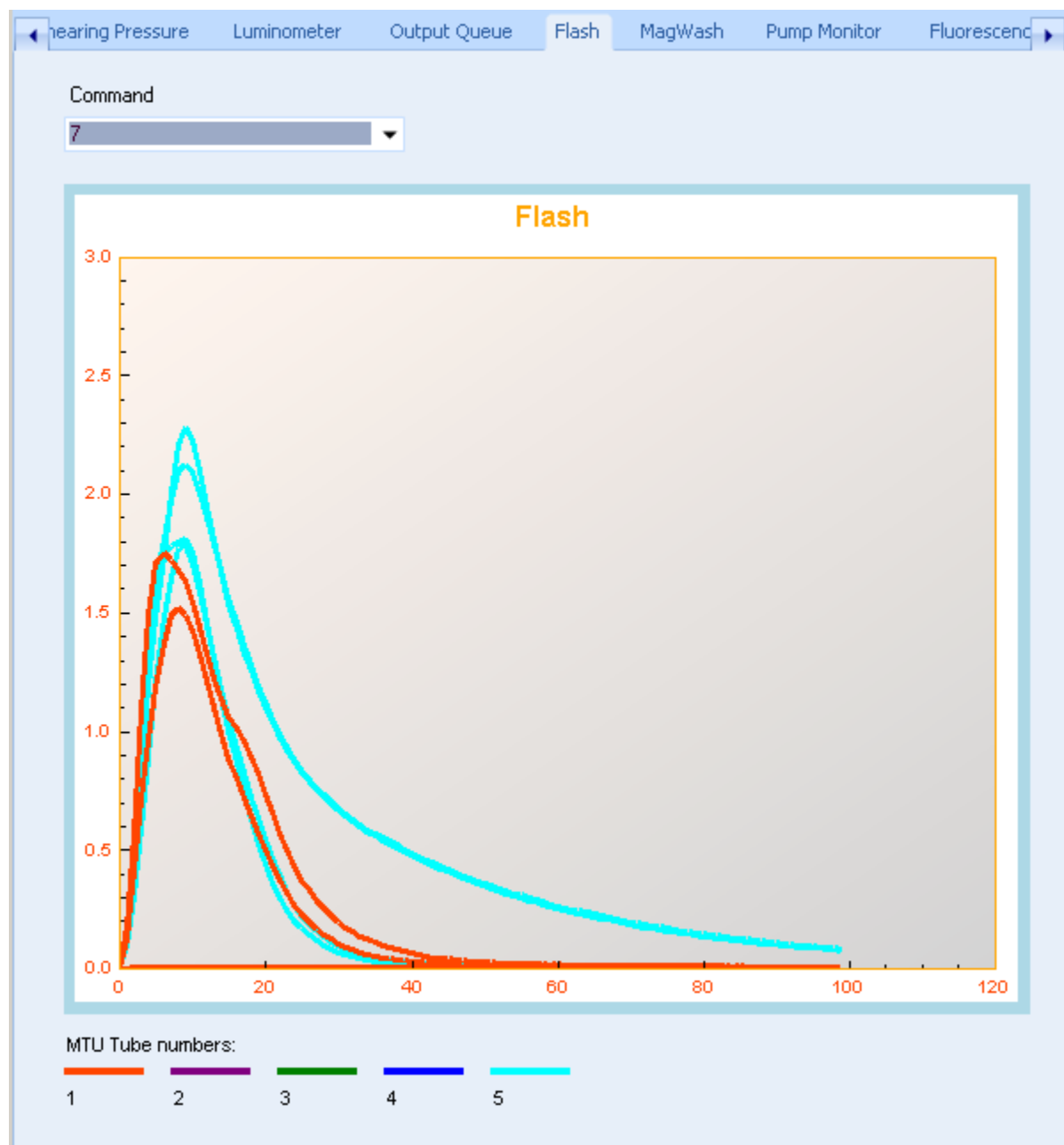
Luminometer



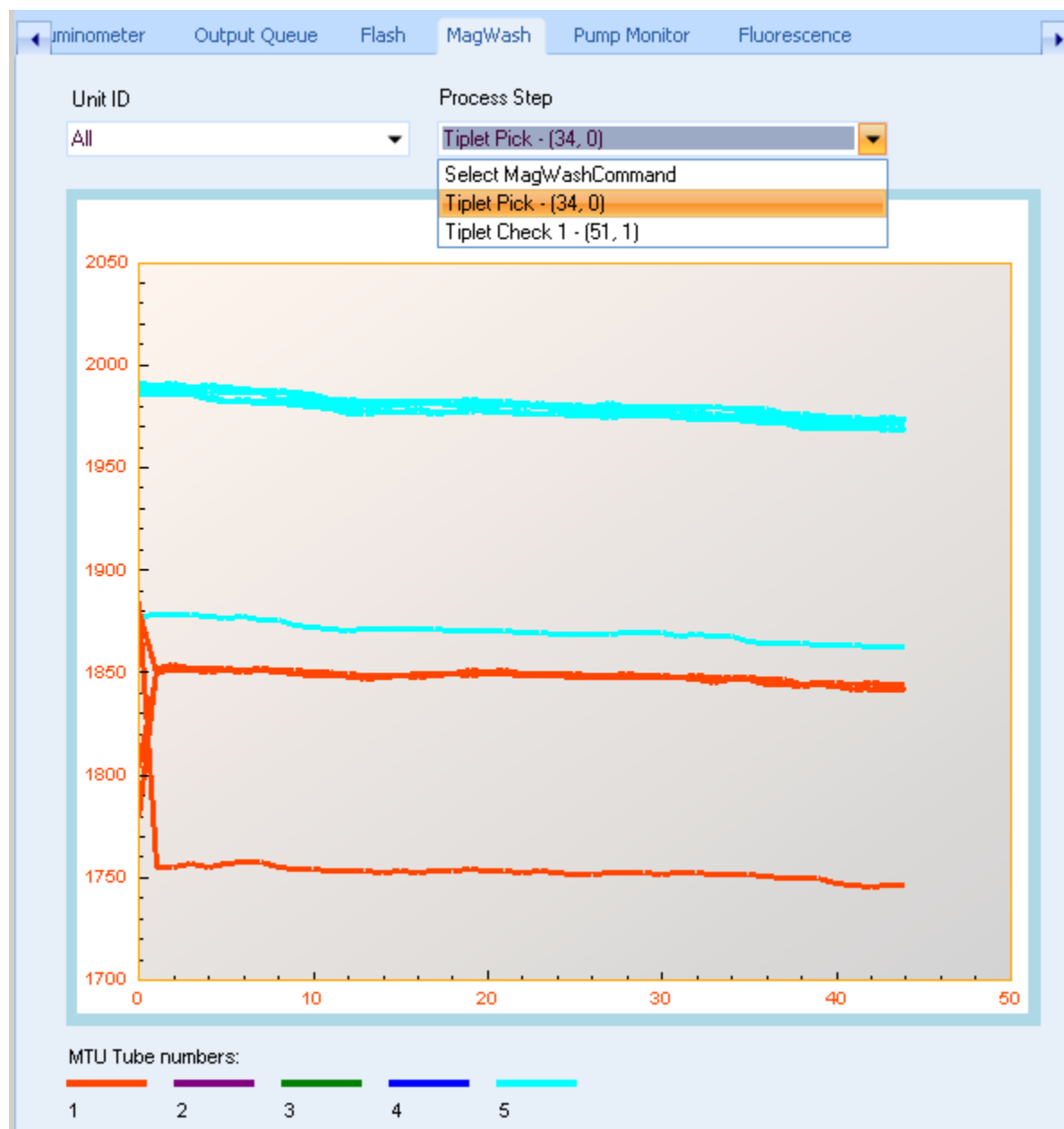
Output Queue



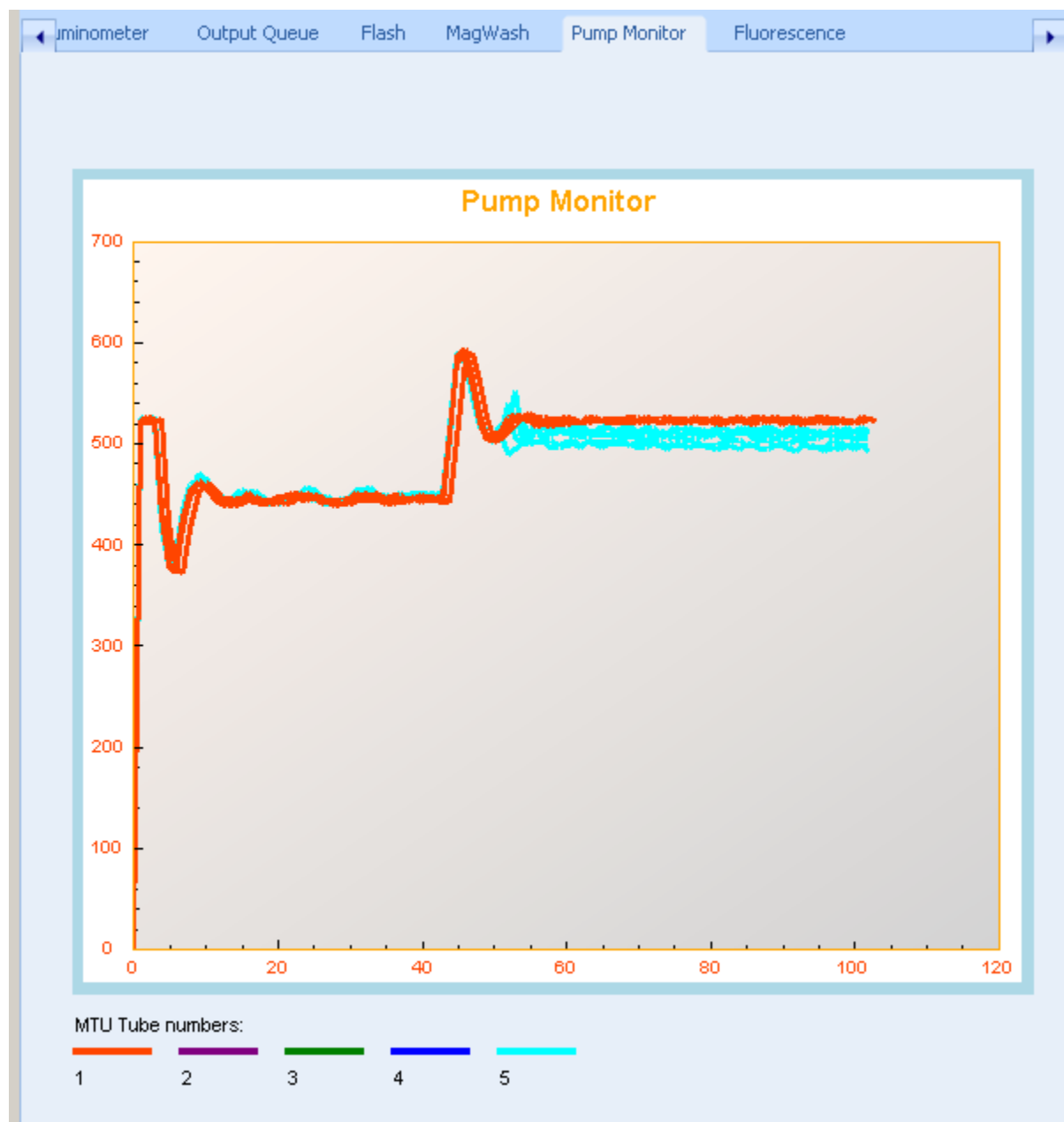
Flash



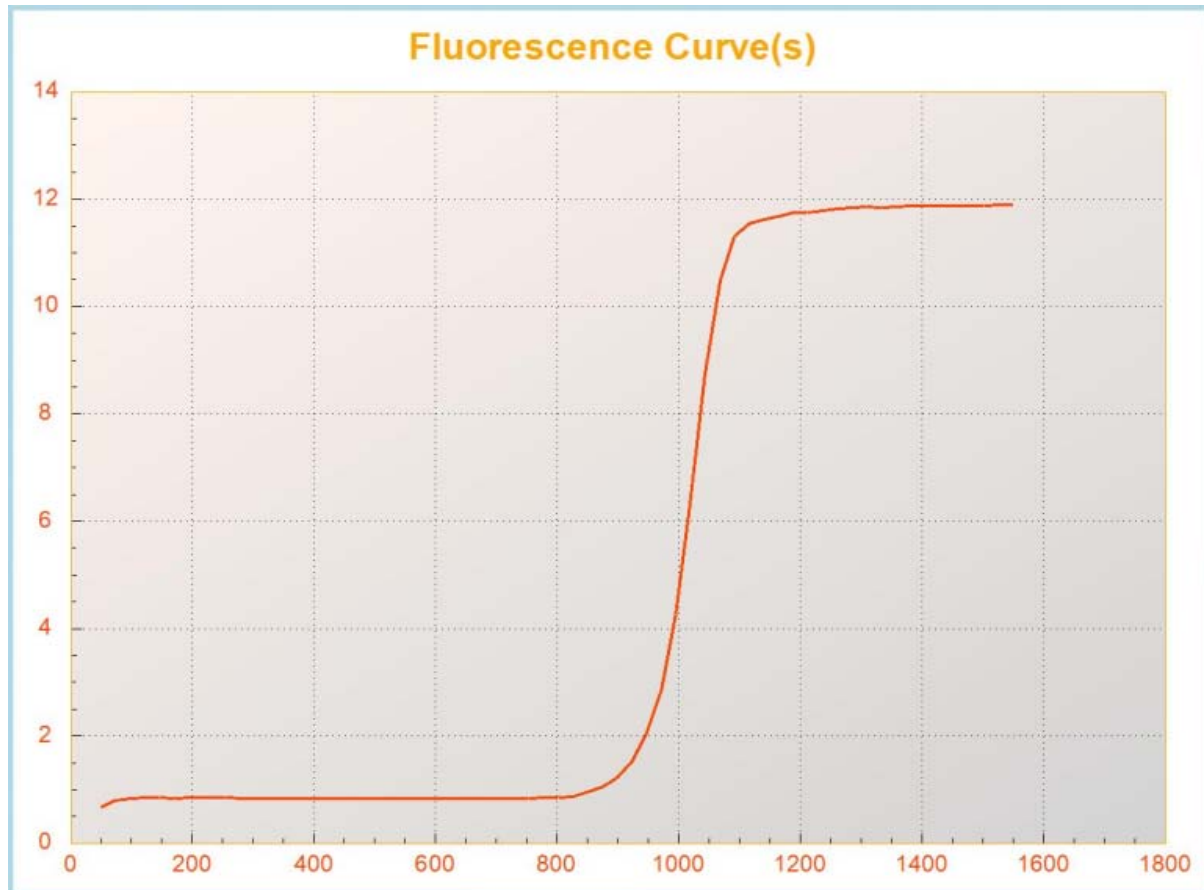
MagWash



Pump Monitor



Fluorescence



APPENDIX C – LIST OF FLAGS

CPF = 7, // Cap pierce failure

FDF1 = 8, // SRS-EEH-263 The system shall apply an FDF1 result flag to the affected tests if a failure is detected during oil dispense.

FDF2 = 9, //

FDF3 = 10, // SRS-EEH-949 The system shall apply a FDF3 result flag to the affected tests if a failure occurs during auto detect I dispense.

FDF4 = 11, // SRS-EEH-950 The system shall apply a FDF4 result flag to the affected tests if a failure occurs during auto detect II dispense.

HB = 12, // SRS-EEH-283 The system shall apply an HB result flag to the affected tests if the luminometer fails the background check operation.

IUO = 14, // For Investigational Use Only

LPF = 15, // Luminometer positioning failure

LDF = 16, // Luminometer read delay failure. This indicates read sequence timings where not within allowed tolerances (Detect 2 delay or PMT delay)

M = 17, // Sample ID of a new test request was manually entered via the keyboard when a barcode was not readable by the instrument.

PMFT = 18, // SRS-EEH-935 The system shall apply a PMFT flag to the affected results when the sample pipettor arm has an unrecoverable hardware error during TCR/TER pipetting.

PMFS = 19, // SRS-EEH-936 The system shall apply a PMFS flag to the affected results when the sample pipettor arm has an unrecoverable hardware error during sample pipetting.

ML1 = 20, // SRS-EEH-427 The system shall apply a ML1 result flag to the affected tests if the residual volume as defined in the Assay SRS is exceeded after TCR/Sample has been aspirated.

ML2 = 21, // SRS-EEH-977 The system shall apply a ML2 result flag to the affected tests if the detected fluid height is above a system configurable limit after wash buffer dispense.

ML3 = 22, // SRS-EEH-977 The system shall apply a ML2 result flag to the affected tests if the detected fluid height is above a system configurable limit after wash buffer dispense.

ML5 = 24, // SRS-EEH-980 The system shall apply an ML5 result flag to the affected tests if the residual volume as defined by the PRD is exceeded after the wash buffer has been aspirated on the last wash.

MT1 = 25, // SRS-EEH-958 The system shall apply an MT1 result flag to the affected tests if 1 or more triplets are not present on the aspirators at the start of the Mag Wash operation.

NTA = 27

NTI = 28

PFS = 29, // SRS-EEH-937 The system shall apply a PFS flag to the affected results when a pump failure occurs during specimen pipetting.

PFT = 30, // SRS-EEH-938 The system shall apply a PFT flag to the affected results when a pump failure occurs during TCR/TER pipetting.

RAFA = 31, // Reagent aspiration failure during aspiration of Amplification reagent.

RAFE = 32, // Reagent aspiration failure during aspiration of Enzyme reagent.

RAFP = 33, // Reagent aspiration failure during aspiration of Probe reagent.

RAFSL = 34, // Reagent aspiration failure during aspiration of Select reagent.

RAFT = 35, // Reagent aspiration failure during aspiration of TCR reagent.

RAFS = 36, // Reagent aspiration failure during aspiration of sample.

RDFA = 37, // SRS EEH-981 The system shall apply a RDFA flag to affected tests when an amplification reagent dispense verification fails.

RDFE = 38, // SRS EEH-982 The system shall apply a RDFE flag to affected tests when an enzyme reagent dispense verification fails.

RDFP = 39, // SRS EEH-983 The system shall apply a RDFP flag to affected tests when a probe reagent dispense verification fails.

RDFSL = 40, // SRS EEH-984 The system shall apply a RDFSL flag to affected tests when a selection reagent dispense verification fails.

RDFT = 41, // SRS EEH-985 The system shall apply a RDFT flag to affected tests when a TCR dispense verification fails.

RDFS = 42, // RDV indicated a failure when dispensing sample

S = 45

SID = 46

PTF = 48, // SRS-EEH-385 The system shall apply a PTF result flag to the affected tests if the pipettor arm is unable to pick a tip.

IT1 = 49, // SRS-EEH-368 The system shall apply an IT1 result flag to the affected tests present in incubator 1 if the temperature is out of range.

IT2 = 50, // SRS-EEH-955 The system shall apply an IT2 result flag to the affected tests present in incubator 2 if the temperature is out of range.

IT3 = 51, // SRS-EEH-956 The system shall apply an IT3 result flag to the affected tests present in incubator 3 if the temperature is out of range.

TC = 52, // SRS-EEH-374 The system shall apply a TC result flag to the affected tests in any MTU present in the chiller if the temperature is out of range

URM2 = 56, // SRS-EEH-454 The system shall apply a URM2 result flag to all of the tests in an MTU with a barcode that is not read correctly before entering the luminometer and the MTU has had tests pipetted to it.

VVFS = 57, // LLS measurement of TCR/Sample in MTU tube falls outside than expected volume

VVFT = 58, // LLS measurement of TCR in MTU tube falls outside (i.e. higher or lower) than expected volume

SW = 59, // SRS-EEH-951 The system shall apply a SW flag when the software is unable to load the data reduction assembly specified in the assay XML file.

// SRS-EEH-952 The system shall apply an SW flag when there is no kinetic data collected for the test and neither an NTO nor an NTI flag was applied to the test.

RVHA = 60, // SRS-EEH-969 The system shall apply an RVHA results flag to the affected tests when the amplification reagent volume check is too high.

RVHE = 61, // SRS-EEH-970 The system shall apply an RVHE results flag to the affected tests when the enzyme reagent volume check is too high.

RVHP = 62, // SRS-EEH-971 The system shall apply an RVHP results flag to the affected tests when the probe reagent volume check is too high.

RVHSL = 63, // SRS-EEH-972 The system shall apply an RVHSL results flag to the affected tests when the selection reagent volume check is too high.

TRC = 64, // SRS-EEH-875 The system shall apply a TRC result flag to the affected tests if a reagent bay temperature excursion is detected.

UMTU = 66, // SRS-EEH-874 The system shall apply a UMTU result flag to the affected tests if it detects that an unexpected MTU has been placed in the Luminometer.

RVHT = 67, // SRS-EEH-964 The system shall apply an RVHT result flag to the affected tests when the TCR volume check is too high.

RVHTR = 68, // SRS-EEH-965 The system shall apply an RVHTR results flag to the affected tests when the TER reagent volume check is too high.

RAFTR = 69, // Reagent aspiration failure during aspiration of TER reagent

RDFTTR = 70, // SRS EEH-986 The system shall apply a RDFTTR flag to affected tests when a TER dispense verification fails.

PCL1 = 71, // ProcessControl check failed for AutoDetect1

PCL2 = 72, // ProcessControl check failed for AutoDetect2

TLS = 74, // SRS-EEH-806 The system shall apply a TLS result flag to the affected tests when the heated load station temperature is out of range.

STL = 75, // SRS-EEH-957 The system shall apply a STL result flag to the affected tests when a sample pipettor tip loss is detected inside the reagent bottle.

RTL = 76, // SRS-EEH-677 The system shall apply a RTL result flag to the affected tests when a reagent pipettor tip loss is detected inside the reagent bottle.

NTP = 77, // SRS-EEH-481 The system shall apply a NTP result flag to the affected tests that failed to receive an assay reagent due to no disposable tips available for pipetting.

QNS = 78, // SRS-EEH-960 If the Pre-Sample aspiration volume check fails, the system shall apply a QNS flag the test to the sample.

CLT = 79, // Clot detected

PFR = 80, // SRS-EEH-948 The system shall apply a PFR flag to the affected results when a pump failure occurs during reagent pipetting.

PMFR = 81, // SRS-EEH-947 The system shall apply a PMFR flag to the affected results when the reagent arm has an unrecoverable hardware error during assay processing.

// PCO removed as per complaint 34866

LWF = 85, // SRS-EEH-880 The system shall apply a LWF result flag to the affected tests if the liquid waste container is full.

RUO = 86, // For Research Use Only

DUO = 87, // For Development Use Only

PEO = 89, // For Performance Evaluation Only

CFM = 90, // User bracket control failure (assay processing flag)

cfa = 91, // User bracket control failure (result processing flag) (lower case because it is applied because of result processing)

IHF1 = 92, // SRS-EEH-793 The system shall apply an IHF1 result flag to the affected tests when the incubator detects a failed heater foil in incubator 1.

IHF2 = 93, // SRS-EEH-941 The system shall apply an IHF2 result flag to the affected tests when the incubator detects a failed heater foil in incubator 2.

IHF3 = 94, // SRS-EEH-942 The system shall apply an IHF3 result flag to the affected tests when the incubator detects a failed heater foil in incubator 3.

RVFT = 95, // SRS-EEH-966 The system shall apply an RVFT flag when the TCR volume level recheck is discrepant with the original TCR volume level check.

RVFTR = 96, // SRS-EEH-967 The system shall apply an RVFTR flag when the TER volume level recheck is discrepant with the original TER volume level check.

RVFA = 97, // SRS-EEH-973 The system shall apply an RVFA flag when the amplification reagent volume level recheck is discrepant with the original amplification reagent volume level check.

RVFE = 98, // SRS-EEH-974 The system shall apply an RVFE flag when the enzyme reagent volume level recheck is discrepant with the original enzyme reagent volume level check.

RVFP = 99, // SRS-EEH-975 The system shall apply an RVFP flag when the probe reagent volume level recheck is discrepant with the original probe reagent volume level check.

RVFSL = 100, // SRS-EEH-976 The system shall apply an RVFSL flag when the selection reagent volume level recheck is discrepant with the original selection reagent volume level check.

LSF = 101, // SRS-EEH-807 The system shall apply a LSF result flag to the affected tests when the load station module detects a failed heater foil.

SRR = 103, // SRS-EEH-871 The system shall apply a SRR result flag to the affected tests if the sample pipetting operation is aborted.

ARR = 105, // SRS-EEH-873 The system shall apply an ARR result flag to the affected tests if the reagent pipetting operation is aborted.

MOS = 106, // SRS-EEH-992 The system shall apply an MOS result flag to the affected tests when a magnetic wash station is taken out of service.

ITF1 = 107, // SRS-EEH-794 The system shall apply an ITF1 result flag to the affected tests when the incubator 1 dual thermistors have exceeded the allowable temperature range between each individual thermistor.

ITF2 = 108, // SRS-EEH-943 The system shall apply an ITF2 result flag to the affected tests when the incubator 2 dual thermistors have exceeded the allowable temperature range between each individual thermistor.

ITF3 = 109, // SRS-EEH-944 The system shall apply an ITF3 result flag to the affected tests when the incubator 3 dual thermistors have exceeded the allowable temperature range between each individual thermistor.

IFF1 = 110, // SRS-EEH-795 The system shall apply an IFF1 result flag to the affected tests when incubator 1 detects the fan RPM is not within the specified range.

IFF2 = 111, // SRS-EEH-945 The system shall apply an IFF2 result flag to the affected tests when incubator 2 detects the fan RPM is not within the specified range.

IFF3 = 112, // SRS-EEH-946 The system shall apply an IFF3 result flag to the affected tests when incubator 3 detects the fan RPM is not within the specified range.

MFM = 113, // SRS-EEH-803 The system shall apply a MFM result flag to the affected tests when the magnet of the magnetic wash station cannot be positioned.

IDF = 114, // SRS-EEH-796 The system shall apply an IDF result flag to the affected tests when the incubator detects the incubator door failed to close.

VACF = 115, // SRS-EEH-808 The system shall apply a VACF result flag to the affected tests when the vacuum system has an unrecoverable failure.

RBEA = 116, // SRS-EEH-1041 The system shall apply an RBEA results flag to the affected tests when the amplification reagent volume check indicates that the amplification reagent bottle is empty.

RBEE = 117, // SRS-EEH-1042 The system shall apply an RBEE results flag to the affected tests when the enzyme reagent volume check indicates that the enzyme reagent bottle is empty.

RBEP = 118, // SRS-EEH-1043 The system shall apply an RBEP results flag to the affected tests when the probe reagent volume check indicates that the probe reagent bottle is empty.

RBESL = 119, // SRS-EEH-1044 The system shall apply an RBESL results flag to the affected tests when the selection reagent volume check indicates that the selection reagent bottle is empty.

RBET = 120, // SRS-EEH-1045 The system shall apply an RBET results flag to the affected tests when the TCR reagent volume check indicates that the TCR bottle is empty.

RBETR = 121, // SRS-EEH-1046 The system shall apply an RBETR results flag to the affected tests when the TER reagent volume check indicates that the TER bottle is empty.

RT = 122, // SRS-SWF-2622 If configured to detect repeat barcodes, the system SHALL apply a RT flag to specimens containing repeat IDs that were accepted by the user.

FL1 = 123, // SRS-EEH The system shall apply an FL1 result flag to the affected tests when the fluorometer has an internal error.

FL2 = 124, // SRS-EEH The system shall apply an FL2 result flag to the affected tests when the fluorometer has a target check error.

FL3 = 125, // SRS-EEH The system shall apply an FL3 result flag to the affected tests when the fluorometer has a voltage check error.

DIL1 = 126, // Assay-SRS-972 When the user selects preconfigured dilution 1 for a sample, the system shall apply a DIL1 flag to the affected results.

DIL2 = 127, // Assay-SRS-973 When the user selects preconfigured dilution 2 for a sample, the system shall apply a DIL2 flag to the affected results.

DIL3 = 128, // Assay-SRS-974 When the user selects preconfigured dilution 3 for a sample, the system shall apply a DIL3 flag to the affected results.

DIL4 = 129, // Assay-SRS-975 When the user selects preconfigured dilution 4 for a sample, the system shall apply a DIL4 flag to the affected results.

DILC = 130, // Assay-SRS-976 When the user selects the custom dilution options, the system shall apply a DILC flag to the affected results.